Investigating Advanced Robotic Assisted Physical Rehabilitation - The Pilot Study

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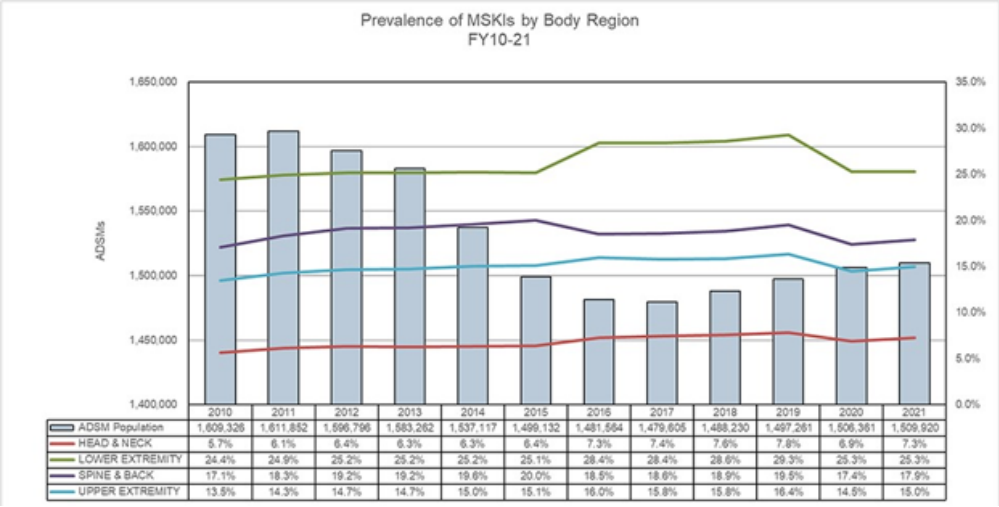
1 RESEARCH BACKGROUND

Military personnel endure intense physical, psychological, and environmental stressors in training and operational environments, making them susceptible to musculoskeletal injuries (MSIs). MSIs are among the most common health issues in the United States Armed Forces, accounting for a substantial share of noncombat-related injuries, limited duty, and disability, thereby degrading U.S. military readiness and imposing substantial costs on the Department of Defense (DoD). A recent 12-year review found that, on average, more than one million active-duty service members experience an MSI each year. Lower-extremity MSIs were most prevalent (24–29%), followed by spine injuries (17–20%) and upper-extremity injuries (14–16%) [1]. Because of the high injury burden, the U.S. military incurs hundreds of millions of dollars annually in private-sector healthcare costs to treat MSIs [1, 2, 3]. There is a need to improve the efficiency of rehabilitation and reduce costs.



	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
ADSM Population	1,609,328	1,611,852	1,596,796	1,583,262	1,537,117	1,499,132	1,481,564	1,479,605	1,488,230	1,497,261	1,506,361	1,509,920
HEAD & NECK	5.7%	6.1%	6.4%	6.3%	6.3%	6.4%	7.3%	7.4%	7.6%	7.8%	6.9%	7.3%
LOWER EXTREMITY	24.4%	24.9%	25.2%	25.2%	25.2%	25.1%	28.4%	28.4%	28.6%	29.3%	25.3%	25.3%
SPINE & BACK	17.1%	18.3%	19.2%	19.2%	19.6%	20.0%	18.5%	18.6%	18.9%	19.5%	17.4%	17.9%
UPPER EXTREMITY	13.5%	14.3%	14.7%	14.7%	15.0%	15.1%	16.0%	15.8%	15.8%	16.4%	14.5%	15.0%

Figure 1: Annual active-duty population and musculoskeletal injury prevalence by body region, fiscal year 2010 to 2021. Graph from Figure 1 in Pav et al., November 2024.

Physical therapy aims to restore lost function in patients with MSIs or other conditions. Across MSIs, two common rehabilitation tasks are passive range-of-motion (PROM) assessment and manual resistance exercises (MRE). In a PROM assessment, the patient relaxes the muscles around a joint while a clinician or robot supports and moves the distal limb segment through the joint's available range of motion; PROM is routine in physical-therapy evaluations and is a central element of early postoperative rehabilitation for many MSIs [4, 5]. During an MRE, the participant assumes a target posture while light resistance is applied to a body segment along a specified direction and gradually increased. The participant may be asked to hold the position, slowly yield to the force, or slowly move against it. Like PROM, MREs are common in outpatient physical therapy for many MSIs [6, 7].

Veterans and civilians in rural areas often travel long distances to reach the nearest therapy clinic — sometimes fewer than 20 miles and sometimes several hundred — depending on

the level of care required [8, 9]. Instead of commuting, many patients can receive physical therapy remotely (telerehabilitation). For veterans and active-duty service members, programs such as VA Video Connect are in use or under development [10, 11]. In the private sector, several programs and devices are commercially available (e.g., CareHand, GenuSport Knee Trainer) [12, 13, 14]. However, most remote-therapy offerings are designed around a single MSI or a single exercise, leaving a market and research gap for a deployable robotic system that can support a broader repertoire of exercises and treat multiple MSIs.

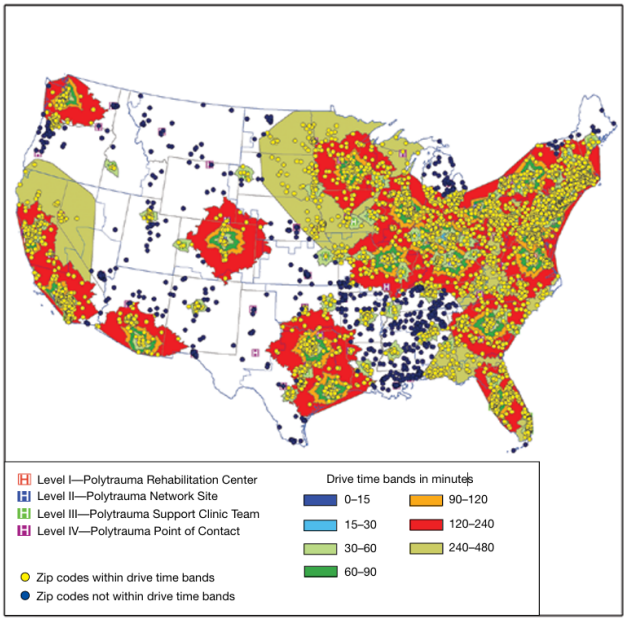


Figure 2: Drive time distance to nearest polytrauma facility for veterans of Operation Enduring Freedom and Operation Iraqi Freedom. Graph from Figure 2 in Ripley et al., October 2009.

Figure 3: Pictures of the exercise program. CareHand app (left), from figure 1 in Sánchez-Laulhe et al., April 2022. Demonstration of the GenuSport Knee trainer (right), from figure 2 in Bäcker et al., February 2021.

Multiple research groups have investigated safe human-robot interaction [15, 16, 17] and physical therapy robots [18, 19, 20, 21, 22]. In this context, the term physical therapy robot refers to an actuated medical device intended to interact with patients by performing rehabilitation, assessment, compensation, or alleviation (RACA). While conventional systems typically require the user to be physically strapped to or manually grasp the physical therapy robot, a nascent field of research explores robotic grippers capable of autonomous, safe manipulation of human limbs [23, 24, 25, 26].

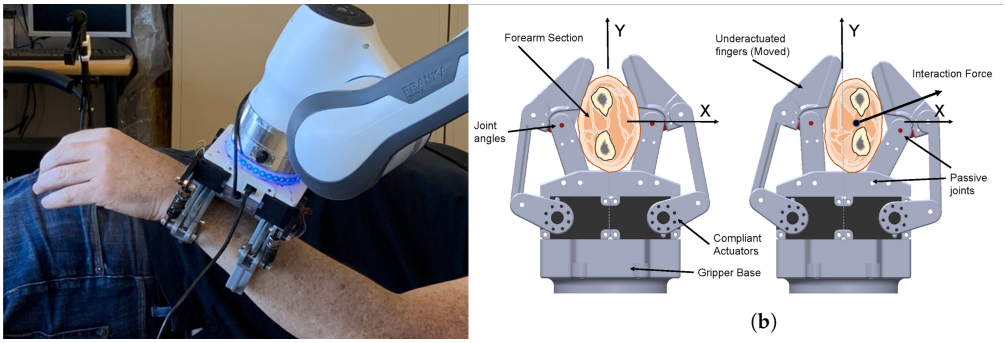


Figure 4: Demonstration of a compliant five-bar grippers for limb manipulation. The left figure is from figure 1 in Ruiz-Ruiz et al. December 2022 and the figure on the right is from figure 1b in Ballesteros et al., May 2020.

1.1 PRIOR RESEARCH

General purpose robotic physical therapy robots utilize grippers engineered for collaborative physical human–robot interaction (pHRI), a domain where the robot and human engage in direct, intentional contact to complete a task [27]. Unlike coexistent or cooperative scenarios, collaborative pHRI is subject to more rigorous safety requirements, as outlined in ISO 15066 and IEC 80601-2-78 [28, 29]. To satisfy these standards, these grippers typically incorporate mechanical passive elements and implement control strategies designed to emulate passive behavior. Modern literature demonstrates the efficacy of such grippers in manipulating peoples' lower arms and wrists. Table 1 presents a comparison between the relevant grippers that were reviewed and the gripper presented in this work.

As demonstrated by Table 1, compliant and/or underactuated five-bar mechanisms are common choices for grippers intended to manipulate human limbs. Compliance and underactuation are well-established characteristics for enhancing adaptability and grasp force observability in robotic end-effectors. These characteristics allow these grippers to passively adapt the shape of the human limb and safely interact with some sort of grasp force feedback.

However, there are some failings in the prior work that are addressed in this work. In Ruiz-Ruiz et al., the authors developed a rigid underactuated gripper with a series elastic actuator (SEA) [23]. Their gripper is the most similar to the gripper presented in this work. Their force estimation model assumed that the limb being grasped has a circular cross-section, that the fingers of the gripper have no thickness, and that the contact points are at

Table 1: Breakdown of relevant prior research findings. SVR stands for support vector regression. P, I, and D stand for proportional, integral, and derivative respectively. These are only the grasp forces that were disseminated by the research studies. The grasp forces were calculated by taking the Euclidean norm of the x- and y-direction contact forces.

Prior Research	**Gripper Mechanism**	**Control Method**	**Grasp Force Capability***	**Intended Anatomy**
Ruiz-Ruiz et al. (2022) [23]	Five-Bar Linkage, Underactuated, Linear SEA	PI Control, Force Estimator	27 ± 2 N	Lower Arm
Lara et al. (2024) [24]	Five-Bar Linkage, Underactuated, Tactile Sensors	PID Control	N/A	Lower Arm
Ballesteros et al. (2020) [25]	Five-Bar Linkage, Underactuated	P Control, SVR Force Estimator	20 ± 5 N	Lower Arm
Hua et al. (2022) [30]	Five-Bar Linkage, Underactuated, Linear SEA	PI Control, Feedforward, Force Estimator	8 ± 1 N	N/A
ANTERO Gripper	Five-Bar Linkage, Underactuated, Linear SEA	Admittance Control, Force Estimator	200 N	Arms and Legs

the midpoint along the phalange pads. These assumptions have been relaxed in the model presented in this work.

In Lara et al., a rigid linkage gripper (without an SEA) was developed with tactile pressure sensing. The grasp forces that this sensorized gripper can produce were not disseminated [24]. In Ballesteros et al., another rigid underactuated gripper without an SEA was developed [25]. This gripper relied on current estimation from the actuator to estimate the contact forces via support vector regression (SVR).

The disseminated grasp forces of these prior grippers produce are less than 30 N. However, the required interaction forces during human-limb manipulation have been shown to exceed 30 N when manipulating a simulated human lower arm at certain arm positions [26]. Furthermore, in physical rehabilitation, more than 100 N may need to be transferred to the patient - in some cases simply to support the weight of a patient's limb, let alone to apply additional resistance during MREs. For example, anthropometric studies that disseminate body-segment parameters indicate that a single human leg can represent 16–18% of total body mass (weight of 120–140 N in a 75 kg individual), while an arm represents around 5% of body mass (weight of 35–40 N) [31, 32].

1.2 GENERAL UPDATES

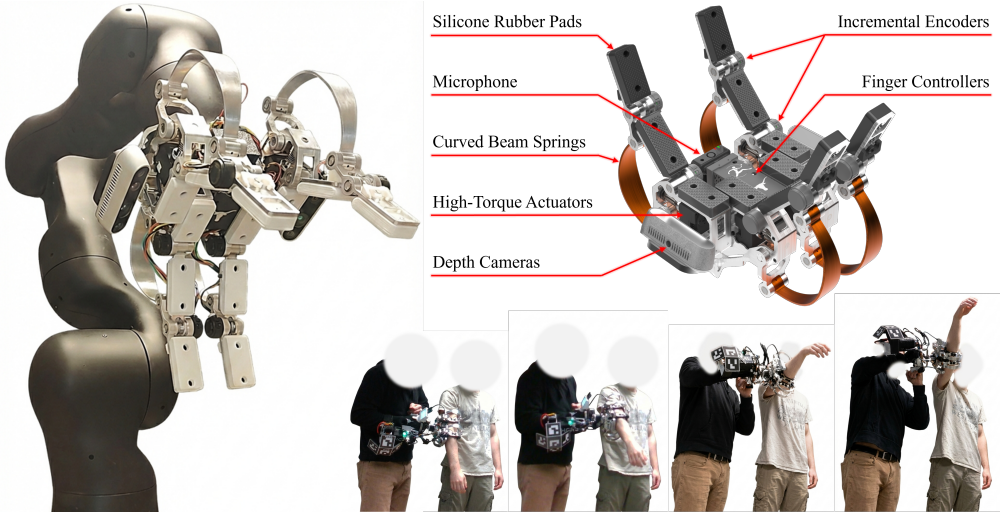


Figure 5: Demonstration of the robotic devices investigated in this work. On the upper right, a diagram of the ANTERO Gripper is shown. On the lower right, the Wearable Data Collection Tool being used during phase 1 of the pilot study is shown. On the left, the Optimo manipulator used in phase 2 of the pilot study is shown.

Researchers in the Human-Centered Robotics Lab have finalized the design of the ANTERO Gripper and implemented a prototype of the Advanced Robotic Physical Therapy System (ARPTS), as shown in Figure 5. A pilot study was conducted to assess the usability

and safety of the ARPTS with able-bodied individuals. While the pilot study results demonstrate that the system is mostly usable and safe, several critical areas require addressing prior to initiating clinical trials. Specific improvements to the ARPTS have been identified for future implementation.

The ARPTS pilot study consisted of two phases: (1) clinician demonstration collection and (2) robot inference. These phases utilized three primary hardware subsystems: the Adaptive Normal-force Tracking and Ergonomic RObotic (ANTERO) Gripper, a wearable data collection tool, and the Optimo robotic manipulator. In addition to these hardware components, a high-level software pipeline was developed using the provided Optimo SDK and a ROS 2 interface. Figure 6 illustrates the subsystem interactions and data flow across both phases of the completed pilot study.

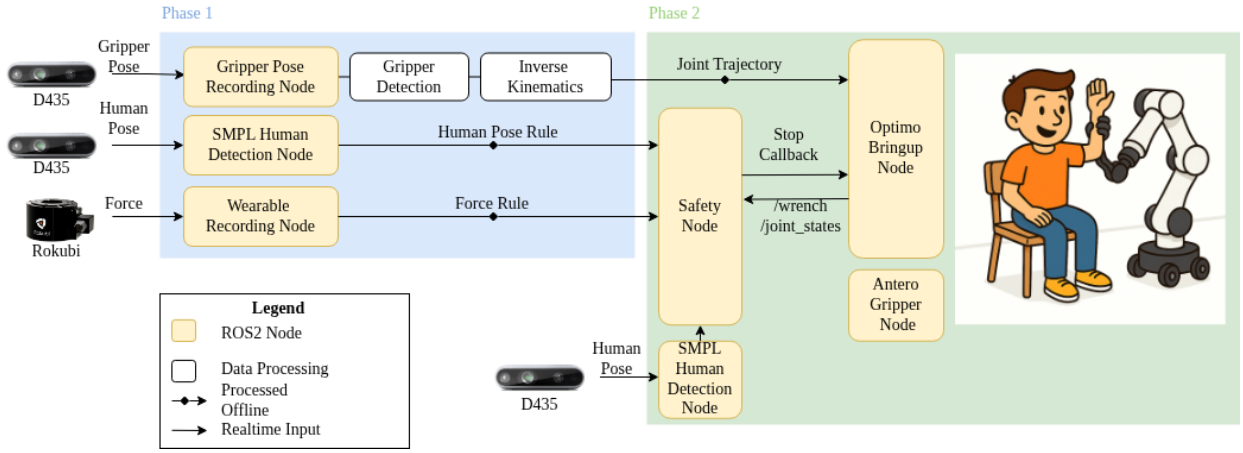


Figure 6: Workflow among ARPTS subsystems

In Phase 1 (demonstration collection), the ANTERO Gripper was mounted on the wearable device, which was operated by a clinician to physically interact with a patient's limbs. The clinician directly controlled the position of the participant's limb and the force transmitted to the participant during these demonstrations. Two third-person Intel RealSense D435i cameras were used to record the gripper motion (gripper pose) and estimate the participants joint positions (human pose). These visual streams are processed through a camera recording node, followed by gripper detection and human pose estimation modules. The resulting outputs were used to generate the Optimo joint trajectories and a human pose rule set. In parallel, a Rokubi 6-DoF force sensor mounted on the wearable recorded the interaction forces, which were processed into a force rule set. These artifacts—joint trajectory, human pose rules, and force rules—served as inputs for Phase 2.

In Phase 2 (robot inference), the ANTERO Gripper was mounted on the Optimo robotic manipulator to reproduce the demonstrated interactions. The recorded joint trajectory from Phase 1 was executed through the *Optimo Bringup Node* to drive the robot motion. A dedicated Safety ROS 2 node continuously monitored execution by comparing real-time observations—including human pose from an external D435 camera and external wrench

and joint states from the robot—against the precomputed human pose and force rule sets. If any safety constraint was violated, the safety node issued a stop callback to halt the robot. Additionally, an ANTERO Gripper node was used to control the gripper during execution, ensuring consistent force and compliance behavior as demonstrated in phase 1. During both sessions, the clinician and researcher were able to override the autonomous controller and release the ANTERO Gripper, if an unexpected event occurred.

2 THE ADVANCED ROBOTIC PHYSICAL THERAPY SYSTEM: OVERVIEW

The pilot study for the ARPTS consisted of two phases: (1) clinician demonstration collection and (2) robot inference. These phases leverage three hardware subsystems: the Adaptive Normal-force Tracking and Ergonomic RObotic (ANTERO) Gripper, a wearable data collection tool, and the Optimo robotic manipulator. On top of these components, we develop a high-level software pipeline using the provided SDK and ROS 2 interface. This section will briefly describe the subsystems and their interconnections.

2.1 THE ANTERO GRIPPER

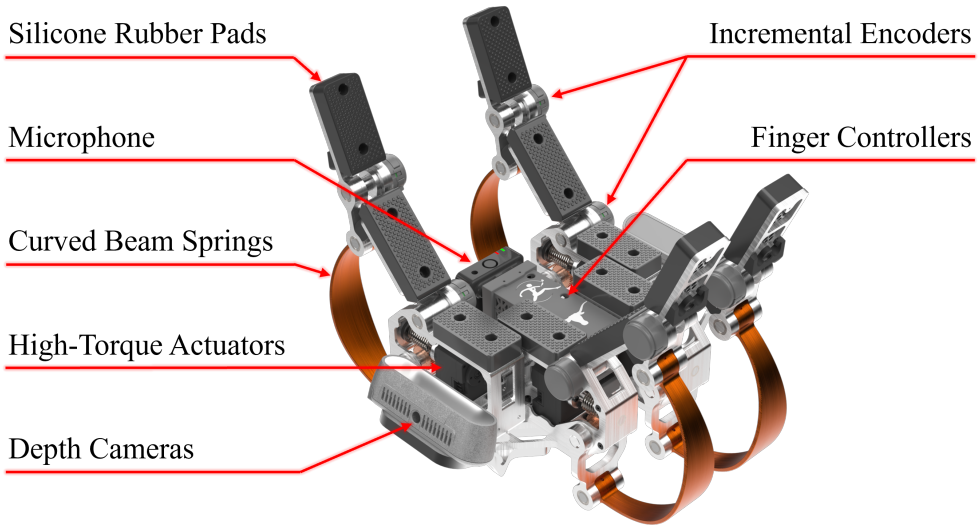


Figure 7: Diagram of the ANTERO Gripper. This design utilizes series elastic actuation to provide inherent compliance and integrated grasp force sensing. Its development yielded several improvements over current state-of-the-art compliant five-bar grippers.

The Adaptive Normal-force Tracking Ergonomic RObotic (ANTERO) Gripper, as shown in Figure 7, is an optimized embodiment of a compliant five-bar mechanism gripper. Because

this gripper is designed to manipulate human limbs, its workspace and grip strength requirements are dictated by human anthropometrics. Measurements regarding human wrist, arm, and leg circumferences were synthesized from the literature [33, 34, 35, 36, 37]. This analysis established a target limb size range from the average wrist circumference (around 150 mm) to the average lower thigh circumference (around 510 mm). For reference, these circumferences correspond to circular cross-sections approximately 48 mm and 162 mm in diameter, respectively. Additionally, because human limbs can weigh upwards of 140 N [31, 32], the ANTERO Gripper is designed to support 200 N of force applied in the direction normal to the palm of the gripper.

The ANTERO Gripper has four symmetrical fingers (ANTERO Fingers) as illustrated in Figure 7. The ANTERO Fingers move and transmit force through a compliant five-bar mechanism with a custom-optimized beam spring that replaces link c. For more information on the beam spring design, refer to the open source repository for the ANTERO Gripper (https://github.com/aambrose3/ANTERO-Gripper.git). Torque applied to the crank (link d) generates a force through the series elastic element, which is subsequently distributed between the proximal and distal finger pads at the contact interface.

Besides the series elastic element, there is a hand-wound torsional spring that provides a restoration torque between the proximal and the distal finger link (link a and link b, respectively) about O_2. This torsion spring is hand-wound from 304 stainless steel cold-drawn wire with a wire diameter of 0.79 mm. The spring has approximately 3.75 active coils and a mean diameter of 11.4 mm. The corresponding spring rate of this spring (K_2) is approximately 0.025 N·m/rad [38]. The spring was wound to achieve a resting angle of approximately -20°.

Each ANTERO Finger utilizes a single motor (Dynamixel XC330-T181-T) located in the palm exoskeleton to drive the crank (link d) of the compliant five-bar mechanism. The actuator incorporates a 15:1 worm drive reduction, significantly enhancing the torque density of the actuator. This integration allows for a relatively small and lightweight motor to be selected, thereby reducing the overall actuator mass. The self-locking characteristic of the worm drive enables the system to support static loads with minimal electrical power consumption. However, these advantages are balanced by trade-offs, specifically low torque transparency and a significant reduction in operational speed.

The no-load speed of the crank is about 0.73 rad/s ($\approx$ 7 RPM). The total mass of the actuator is about 118 g (including the palm exoskeleton), and the actuator can theoretically continuously output 4.8 N·m at 0.25 rad/s while consuming less than 7 W of electrical power. Three encoders are implemented to provide feedback on the finger posture and compute the contact forces as described in [Modeling]. The motor has an absolute magnetic encoder on its output. Through the 15:1 worm drive reduction, ideally, this encoder can read the angle of the crank link with a resolution of 21 arcseconds. Two optical incremental encoders (US Digital E4T-720-250-S-D-M-B) are implemented to read the angles of the proximal finger link (O_1) and distal finger link (O_2). The optical encoders have a resolution of 7.5 arcminutes.

The ANTERO Gripper relies on a hierarchical control structure. An agnostic high-level controller issues instruction packets containing the desired high-level behavior of the

ANTERO Gripper via a Controller Area Network (CAN 2.0 A) bus operating at 250 kbps. The instruction packets are executed by the mid-level controllers, one for each of the four ANTERO Fingers. These instruction packets contain the gripper state (enabled or disabled), commands for the grasp force, and the desired emulated compliance of the compliant five-bar mechanism. The motor controller commands to achieve this behavior are computed and sent to the low-level motor controllers at regular intervals.

The printed circuit board schematic for this custom mid-level controller is shown in Figure 8. All design files and firmware are available in the open-source repository (https://github.com/aambrose3/A Gripper.git).

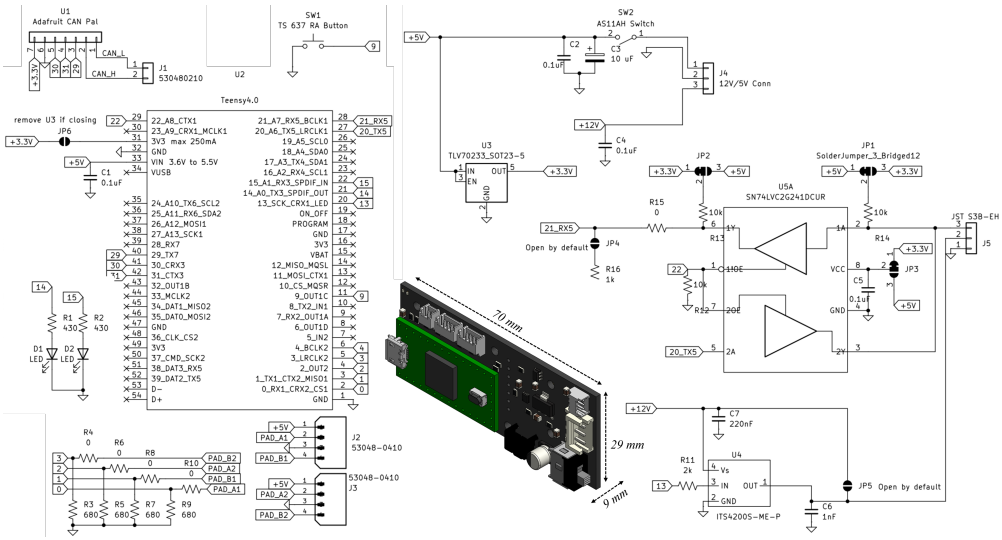


Figure 8: Circuit diagram and rendering of the finger control boards. Each finger has one of these boards that acts as mid-level controllers between an agnostic high-level controller and the low-level motor controllers.

Beyond regulating the grasp force, the ANTERO Gripper has two front facing depth cameras (Intel D435i) and a wireless microphone (RØDE Wireless Go II). These sensors are included to provide a high-level controller with information about the human-robot interaction. For example, the depth cameras with integrated inertia measurement units (IMUs) could be used to autonomously guide the robot manipulator to the patient and close the loop around the physical therapy exercise motion. Additionally, the microphone in conjunction with a speech-to-text algorithm could be used to capture patients' dialogue. This could improve the safety of the high-level controller and the human-robot interaction. Both of these examples are left for future work. The total mass of the ANTERO Gripper (including wires, silicone finger pads, cameras, and the microphone) is approximately 1.54 kg.

Figure 9 illustrates the closed-loop control architecture for each ANTERO Finger. The scheme consists of an admittance controller cascaded with a low-level motor velocity control loop. This allows the desired net grasp force and the desired passive behavior of the compliant

five-bar mechanism to be controlled.

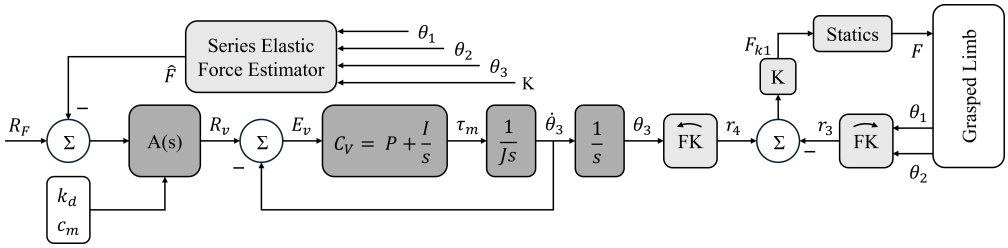


Figure 9: Control diagram for the velocity-sourced collocated admittance controller, A(s), implemented in the ANTERO Gripper embodiment. The dark gray blocks are linear while the light gray blocks are nonlinear. R_F is the desired net grasp force and $\hat{F}$ is the estimated net grasp force. The motor controller is a PI velocity controller that directly interacts with the effective inertia (J) of the actuator and the five-bar mechanism. Here, FK stands for forward kinematics, and the arrows indicate the direction around the closed loop chain.

The grasp force estimator discussed in previous reports has been implemented on the mid-level controllers and is represented by the force estimator block. The admittance controller renders a visco-elastic behavior described by Equation 1, derived from strategies outlined from prior work [39].

$$A(s) = \frac{c_v J_v k - k_v s^2 + k_v J_v k s + c_v k_v k}{c_v J_v k_v k s} \tag{1}$$

In Equation 1, s is the Laplace operator, k_v is the desired virtual stiffness and c_v is the desired emulated motor damping, and J_v is the desired virtual inertia. This emulated/virtual behavior, along with the desired offset grasp force (R_F), is prescribed by the agnostic high-level controller. Unlike prior work with series SEAs and admittance controllers [39, 40, 41], the force produced by the series elastic element F_{k1} is distributed between two contacts to generate a net contact force (F). This force differential results in the emulated stiffness, damping, and inertia being abstracted from the admittance at the contact interface. While the admittance controller can still affect the contact interface admittance, the complex relationship between the emulated admittance parameters and the contact interface admittance, along with extensive stability and passivity criterion, has not been investigated in this work.

2.2 THE WEARABLE DATA COLLECTION TOOL

The wearable data collection tool (WDCT), shown in Figure 10, is used to gather the ground-truth demonstrations in phase 1 of the pilot study. The therapist interacts with the patient through the WDCT and the ANTERO Gripper. The wearable houses a Raspberry Pi 5 and various sensors to record the therapist-patient interaction. The handle on the WDCT has three buttons to operate the wearable and ANTERO Gripper system. The first button (labelled E-Stop in Figure 6) is used to arm the wearable device (which starts and stops recording data). The bottom trigger on the handle is used to enable the ANTERO Gripper,

causing the gripper to close and grasp the limb with around 20 N of total grasp strength. The top trigger is used to adjust the contact interface admittance and grasp force, which are linearly coupled.

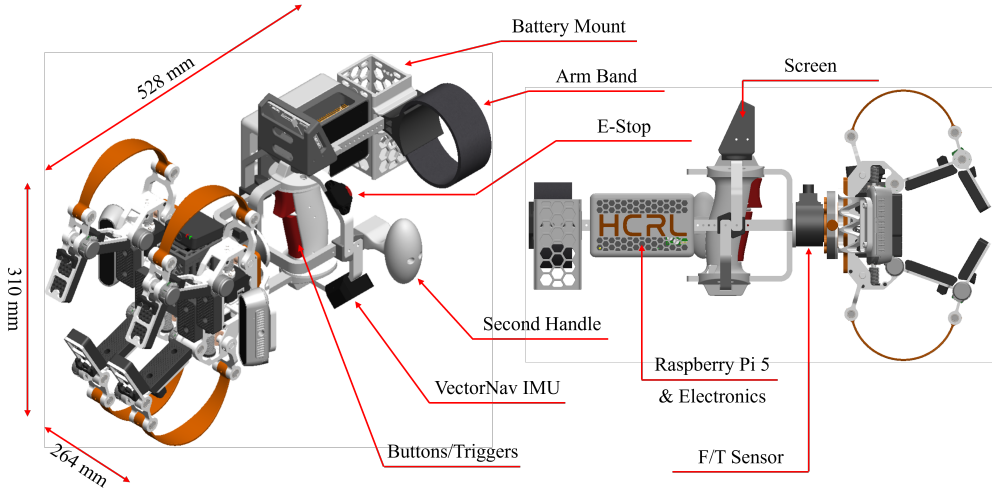


Figure 10: Diagram of the wearable data collection tool used by the clinicians in phase 1 of the pilot study to guide participants through physical therapy-like exercises. This wearable tool is portable, powered by a 22 V battery. The run time for this portable system exceeds 4 hours.

The main wearable computer is the Raspberry Pi 5, that acts like the high-level controller for the gripper and runs ROS2 to interface with the various sensors. Table 2 shows a breakdown of the actions that the Raspberry Pi 5 performs during therapist-patient interactions.

2.3 THE OPTIMO MANIPULATOR

The Optimo robotic manipulator was utilized to manipulate human limbs and reenact physical therapy exercises. A high-level controller was developed atop the existing SDK and ROS2 wrapper. Figure [X] illustrates the interaction between the various subsystems within this controller. In Phase 1, two Intel RealSense D435 cameras captured gripper and human poses, which were processed to generate a joint trajectory file and a human-pose "rulebook" for Phase 2. Simultaneously, a Rokubi 6-DoF force sensor was used to establish force-related rules based on clinician demonstrations.

2.4 THE SOFTWARE PIPELINE

Gripper Pose Recording Node	30 Hz	Inputs : color images, depth images, camera intrinsics	Outputs : rosbag

Table 2: List of the sensors implemented on the wearable and ANTERO Gripper that are sampled during therapist-patient interactions. Here IMU stands for inertial measurement unit.

Sensor	**Sample Rate**	**Communication Protocol**	**Additional Information**
Reads the buttons (Limit Switches)	GPIO Interrupt Service Routine (< 10 bps)	Digital	To arm the wearable device (starts/stops recording and enables the gripper). The lower trigger closes the gripper.
Reads the top trigger (A1454 Hall Effect Sensor)	25 Hz (∼200 bps)	I2C	A MEMS hall-effect-sensor to read the position of the wearable handle's top trigger
Reads the IMU (VectorNav 100s)	400 Hz (∼ 115 Kbps)	Serial USB	Provides 9-DOF information. Linear acceleration, angular rate, and magnetometer data.
Reads the force-torque sensor (BFT-ROKS-SER-M8)	200 Hz (∼ 230 Kbps)	Serial USB	Implemented between the wearable and the ANTERO Gripper, this sensor record the cartesian forces and moments transferred between the patient and the wearable.
Reads the two cameras (Intel RealSense D435i)	400 Hz – IMU (∼ 77 Kbps), 30 Hz - images (∼ 76 Mbps)	Serial USB	Reads the IMUs at 400 Hz and reads color and depth images at 30 Hz. The resolution of the color images is 320x180 pixels and the depth images is 480x270 pixels.
Reads RODE Wireless Go II & 48 kHz at 24-bit (< 2 Mbps)	Serial USB + proprietary wireless communication with 128-bit encryption.	Records audio during the therapist-patient interactions	This microphone was not used in this research study. No audio data was collected during this study.

The *Gripper Pose Recording Node* is a ROS2 nodes that receives and record the color images, depth images and camera intrinsics from the Intel RealSense D435 camera.

SMPL Human Detection Node	10 Hz	Inputs : color images, depth images, camera intrinsics	Outputs : rosbag, key joint locations, safety vector on key joints

The *SMPL Human Detection Node* is a ROS2 node that receives and records color images, depth images, and camera intrinsics from the Intel RealSense D435 camera. It then processes the recorded color and depth images. We first obtain the key joint locations and rotation matrices by applying the Skinned Multi-Person Linear (SMPL) model to the human figures present in the video.

We then represent the direction of the right upper arm in a body-centered coordinate frame. First, we define three orthonormal axes using body joints. The rightward axis is defined as:

$$\mathbf{e}_{\text{right}} = \frac{\mathbf{p}_{\text{r_shoulder}} - \mathbf{p}_{\text{l_shoulder}}}{\|\mathbf{p}_{\text{r_shoulder}} - \mathbf{p}_{\text{l_shoulder}}\|} \tag{2}$$

Here, $p_{r_shoulder}$ represents the pixel coordinates of the right shoulder of the participant and $p_{l_shoulder}$ represents the pixel coordinates of the left shoulder of the participant. The double vertical bars represent the norm of the vector. The upward axis is defined as:

$$\mathbf{e}_{\text{up}} = \frac{\mathbf{p}_{\text{neck}} - \mathbf{p}_{\text{pelvis}}}{\|\mathbf{p}_{\text{neck}} - \mathbf{p}_{\text{pelvis}}\|} \tag{3}$$

Here, p_{neck} represents the pixel coordinates of the center of the neck, a key point from the SMPL model, and p_{pelvis} represents the pixel coordinates of the center of the navel, another key point from the SMPL model. The forward axis is computed using the cross product:

$$\mathbf{e}_{\text{front}} = \frac{\mathbf{e}_{\text{right}} \times \mathbf{e}_{\text{up}}}{\|\mathbf{e}_{\text{right}} \times \mathbf{e}_{\text{up}}\|} \tag{4}$$

The direction of the right upper arm is then defined as a normalized vector:

$$\mathbf{v}_{\text{upper}} = \frac{\mathbf{p}_{\text{r_elbow}} - \mathbf{p}_{\text{r_shoulder}}}{\|\mathbf{p}_{\text{r_elbow}} - \mathbf{p}_{\text{r_shoulder}}\|} \tag{5}$$

Here, p_{r_elbow} represents the pixel coordinates of the elbow on the participants right arm. Finally, we project this vector onto the body-centered coordinate frame to obtain:

$$x = \mathbf{v}_{\text{upper}} \cdot \mathbf{e}_{\text{right}}, \quad y = \mathbf{v}_{\text{upper}} \cdot \mathbf{e}_{\text{up}}, \quad z = \mathbf{v}_{\text{upper}} \cdot \mathbf{e}_{\text{front}} \tag{6}$$

where x, y, and z represent the relative orientation of the upper arm with respect to the torso along the lateral (right), vertical (up), and forward directions, respectively.

Wearable Recording Node	30 Hz	Inputs :	Outputs : rosbag, force

The *Wearable Recording Node* is a ROS2 nodes that receives and record the color images, depth images, camera intrinsics, imu data and the 6-DoF force sensor data from the Intel RealSense D435i cameras, VectorNav 100s, force-torque sensor onboard the wearable.

Gripper Detection	30 Hz	Inputs : color images, depth images	Outputs : gripper trajectory

The *Gripper Detection Module* takes as input synchronized color and depth images from the camera recording node. It first detects the five ArUco markers attached to the wearable device using OpenCV's ArUco detector. For each detected marker, we estimate its pose by combining two sources of information. Specifically, OpenCV's solvePnP is used to recover the in-plane translation (x, y) and yaw angle from the 2D–3D correspondences, while depth measurements at the four marker corners are used to estimate the out-of-plane translation (z) as well as the roll and pitch angles. The resulting marker poses are then transformed into the gripper frame using a predefined rigid transformation between each marker and the gripper. Finally, multiple gripper pose estimates obtained from different markers are fused into a single robust estimate by averaging across valid detections, reducing noise through Exponential Moving Average (EMA).

Inverse Kinematics	Offline	Inputs : gripper trajectory	Outputs : joint trajectory

The *Inverse Kinematics* module takes in the gripper trajectory in the camera frame and transfers it into the initial gripper frame. Then, we obtain the URDF file for the Optimo robot and solve inverse kinematics with the Pink library on top of Pinocchio, using a taskspace QP formulation. The output is a 200Hz joint trajectory that can be directly read and executed by the Optimo.

Safety Node	50 Hz	Inputs : realtime human pose, realtime force, human pose rulebook, force rulebook	Outputs : stop service call to bringup node

The *Safety Node* monitors safety conditions based on both human pose and interaction forces. We first apply the SMPL model to human figures captured from the real-time D435 input to obtain key joint locations, and then compute safety vectors for selected joints. These vectors are compared against the corresponding safety vectors stored in the rulebook collected during Phase 1 through SMPL-based human pose estimation. A tolerance of 10% of the observed range is applied to the minimum and maximum bounds, within which a warning (yellow) is issued. A stricter tolerance of 33% beyond the range is also defined; if exceeded, the system flags an unsafe condition (red) and triggers a stop service to halt the robot. Similarly, the safety node monitors real-time force readings in the x, y, and z

directions at the end effector and compares them against the force rulebook generated from the gripper recording module. The real-time force is calculated by the difference between actual external wrench reading at the run and baseline external wrench reading when there is no payload on the end effector. Any violation of the nominal force range results in a warning. In addition, a fixed tolerance margin of ±20 N beyond the minimum and maximum bounds is enforced; exceeding this threshold triggers an unsafe condition and stops the robot.

Optimo Bringup Node	1000 Hz	Inputs : joint trajectory, stop service call	Outputs : Optimo robot motion; publish robot related topics

The *Optimo Bringup Node* is a ROS2 wrapper on the SDK that operates the Optimo robot. In our study, the bringup node takes in joint trajectory and excute the trajectory on the robot and publishes robot related topics like joint poses, external wrenches, etc. This node also takes in stop service call from the safety node to soft stop the robot if safety conditions are violated.

ANTERO Gripper Node	20 Hz	Inputs : press any key	Outputs : ANTERO gripper open/close

The *ANTERO Gripper Node* opens or closes the ANTERO gripper when any key is pressed.

3 PILOT STUDY METHODOLOGY

The pilot study investigating upper-limb rehabilitation with the ARPTS was designed to demonstrate the safety and preliminary usability of the ARPTS. The following sections details the methods employed for the pilot study.

3.1 STUDY OVERVIEW AND PARTICIPANT INCLUSION

This was a single-site study where all direct participant interactions took place in the Human Centered Robotics Laboratory (HCRL). Each participant was asked to participate in two sessions, but they were not required to attend both. Participants only proceeded to the second session after completing the first session and receiving an invitation back from the research team. Participants were permitted to opt out of the study at any time.

During scheduling, participants were instructed to attend the sessions abiding by the following attire and safety procedures:

- Wearing closed-toe shoes.
- Pulling back long hair (if applicable).
- Wearing athletic clothing that did not obscure joint motions.

If a participant arrived without adhering to these procedures, their session was rescheduled. Usually, these studies require that athletic clothing to be worn to not obscure the human pose estimation algorithms, but for our study we allowed baggy clothing and used a relatively new AI algorithm for detecting the joints positions of the participants. Each of the two sessions did not exceed 1.5 hours (totaling a maximum of 3 hours for those who participated in both).

The subject population included staff, and students at the University of Texas at Austin. However, these populations were not exclusively targeted by this study. This study was available to anyone who saw the recruitment materials.

Participant Inclusion Criteria:

- At least 18 years of age
- Not more than 65 years of age
- Proficient/fluent in speaking English
- Have no neuromuscular conditions that affect their ability to operate their arms, legs, or hands.
- Able to provide informed consent/parental permission

Clinician Inclusion Criteria:

1. The clinician is a licensed or practicing physical therapist.

The needed sample size was determined through a-priori power analysis. A-priori power analysis was conducted to detect a large effect size ($d \geq 2$) with a statistical power over 80% for a difference from a constant t-test. Here the constant is the ground truth gathered during the first phase of the pilot study. For example we will be comparing the force the robotics systems imparts on the human subject in phase 2 compared with the force imparted on the human subject from phase 1. With a 5% chance of committing a type 1 error, the estimated total number of participants needed is 4. Below is the output of the G*Power 3.1.9.7 analysis:

At least two members of the research team proctored each session with every participant (A robotics researcher and a physical therapist). The robotics researcher was responsible for monitoring the state of the robotic system and the physical therapist was responsible for monitoring the participants' well being. During the first session, the clinician wore the data collection tool and gripper and performed physical therapy-like exercises with the participant. The wearable data collection tool recorded data regarding the physical interaction while performing these mock exercises. In phase 2, the Optimo robotic manipulator autonomously interacted with the participants instead of the physical therapist. The participant and robotic system were always under strict supervision by the clinician and researcher.

Table 3: A priori power analysis for one-sample t-test (one-tailed).

Analysis Type	A priori: Compute required sample size
Input Parameters	
Tail(s)	One
Effect size (d)	2
Significance level (α)	0.05
Power ($1 - \beta$)	0.8
Output Results	
Noncentrality parameter (δ)	4.0000000
Critical t	2.3533634
Degrees of freedom (df)	3
Total sample size	4
Actual power	0.9084467

3.2 PRE-STUDY PROCEDURE

A pre-study phase ensured that the robotic system elements (wearable data collection device, robotic gripper, and robotic manipulator) functioned as intended. Each element was tested individually and collectively using a mannequin arm and leg (Simulaids Rescue Randy, 105 lbs) within one day prior to participant arrival. Advertising and scheduling also commenced during this phase.

3.3 PHASE 1 PROCEDURE

Upon the participant's arrival at the laboratory, the researcher introduced the robotic system elements. The researcher completed the informed consent process, answered any questions, and reiterated that the study was voluntary.

During the session, the clinician used the wearable data collection device and robotic gripper to perform and capture data across multiple mock physical therapy exercises. The selected movement exercises were routine components of outpatient physical therapy:

- **Passive Range of Motion (PROM):** The participant relaxed the muscles around a joint while the therapist supported and moved the distal segment through its available range of motion.
- **Manual Resistance Techniques (MRT):** The participant assumed a limb posture while light resistance was applied and gradually increased. The participant statically held the position, slowly yielded to the force, or slowly moved against it.

Participants were asked to stand or to sit in a chair depending on which was more comfortable for the clinician. Before any exercise, participants were informed where the clinician would grasp their limb, and they were offered a demonstration on the mannequin first.

The documented procedure for phase 1 is as follows:

1. Position the two cameras to capture the gripper and human pose.
    (a) Place one camera directly facing the human torso for human pose detection.
    (b) Place the other camera in a non-occluded position to capture the wearable motion throughout the exercise. Make sure there are at least 1 ArUco markers detectable by the camera at any time by performing mock motions before data collection. See Figure 11

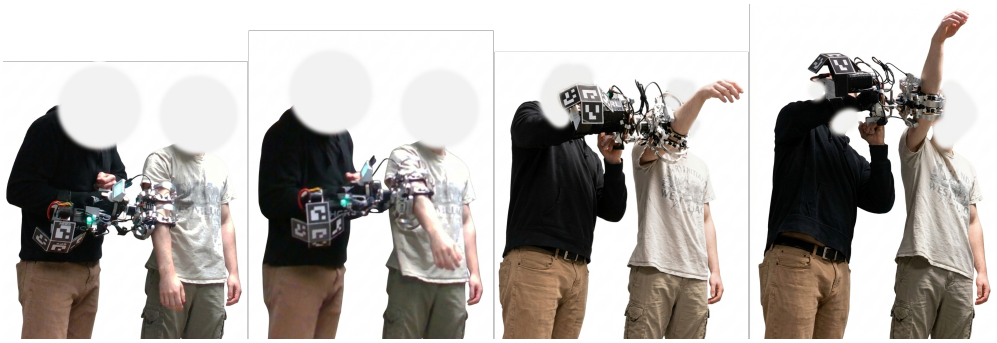


Figure 11: Demonstration of the proper position of the second camera meant for tracking the motion of the wearable device.

2. Boot the wearable device.
    (a) After the homing sequence, remove the safety stops (four of them).
    (b) Wait until the green light on the wearable GUI labeled "gripper connection" to turn green. See Figure 12.

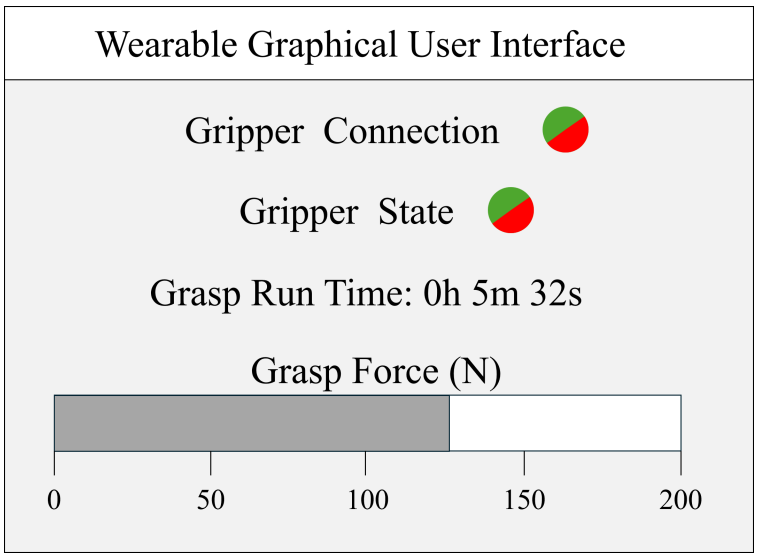


Figure 12: Illustration of the GUI implemented on the wearable data collection tool.

3. The clinician or robotics researcher positions the patient in the correct spot and orients them correctly.
   (a) Verify a test motion of the exercise, checking for gripper tracking and human pose estimation. Should not grasp onto the limb of the participant yet.
4. If tracking look good, the robotic researcher starts recording the gripper camera and informs the clinician to proceed. See the following instructions:
   (a) Download the ROS package from GitHub: https://github.com/jidaxian010/pt_collect.git
   (b) Launch the node:
   `ros2 launch pt_collect pt_collect_launch.py`
   (c) Record and save the rosbag to Downloads:
   `ros2 bag record -a -o ~/Downloads/p8-b2-g`
   (d) Monitor the marker detection and gripper estimation in real time:
   `ros2 run rqt_image_view rqt_image_view /aruco/gripper_pose_four`
5. Immediately following the previous steps, the robotics researcher starts recording the human pose camera.
   (a) Download the ROS package from GitHub:
   https://github.com/jidaxian010/human_pose_vector.git
   (b) Launch the camera and the node:
   `ros2 launch human_pose_vector pose_vector.launch.py`
   `bash ~/ros2_ws/run_pose_node.sh`

(c) Record and save the rosbag to Downloads:
`ros2 bag record -a -o ~/Downloads/p8-b2-h`

6. After 5 seconds, the clinician starts recording on the wearable.

   (a) Press the E-Stop button. The "Gripper State" light should turn green.

7. The robotics researcher should take notes about the perceived success of the task. e.g. the clinician might say "it felt like you trusted me more this time" talking with the participant. This would indicate that the data collected from this set might be a higher quality than previous data.

8. For both of the exercises in this study, the clinician grasps the participant across the elbow on the participants right arm. This was done by pressing and releasing the bottom trigger. Do not hold down this trigger.

9. The clinician performs three repetitions of a therapy-like exercise.

10. After performing three repetitions, the clinician detaches the ANTERO Gripper from the participant by pressing and releasing the lower trigger again. The Gripper State light should turn red.

11. The robotics researcher stops both camera recordings and verified the integrity of the saved bag files.

12. At roughly the same time, the clinician stops the wearable recordings (after fully opened) by pressing the E-stop button.

13. This process (steps 3-11) was repeats three times per exercise.

3.4 PHASE 1 POSTPROCESSING

Immediately following the first session, data was consolidated onto a secure university-owned hard drive. All data was uploaded to UT Box, and the local memory on the wearable device and the secondary computer with the a GPU was wiped. The encrypted primary data repository was strictly preserved and never directly manipulated. Physical documents were stored in a locked cabinet within the key-card-restricted HCRL.

Machine learning policies were developed to allow the robotic manipulator to reproduce the motion and interaction forces generated by the clinician. These control policies were tested in simulation environments (Pinocchio based simulator, Python, and MATLAB) on a secure UT computer, followed by physical testing on the mannequin in the HCRL. The streamlined process for generating these control policies are as follows:

1. Run the *Gripper Pose Recording Node* to generate the optimo joint trajectory and force rule. See Figure 13.

(a) Download the repo from GitHub: https://github.com/jidaxian010/RobotPT.git

(b) Create and activate the conda environment:
```
conda env create -f environment.yaml
conda activate robotpt
```

(c) Select one rosbag for each motion based on participant and clinician feedback (see any notes taken during phase 1 step 7) and generate an MP4 preview:
```
python scripts/preview.py
```

(d) Manually crop out the approach and ending stage. Generate gripper pose in the initial gripper frame: `python scripts/generate_gripper.py`

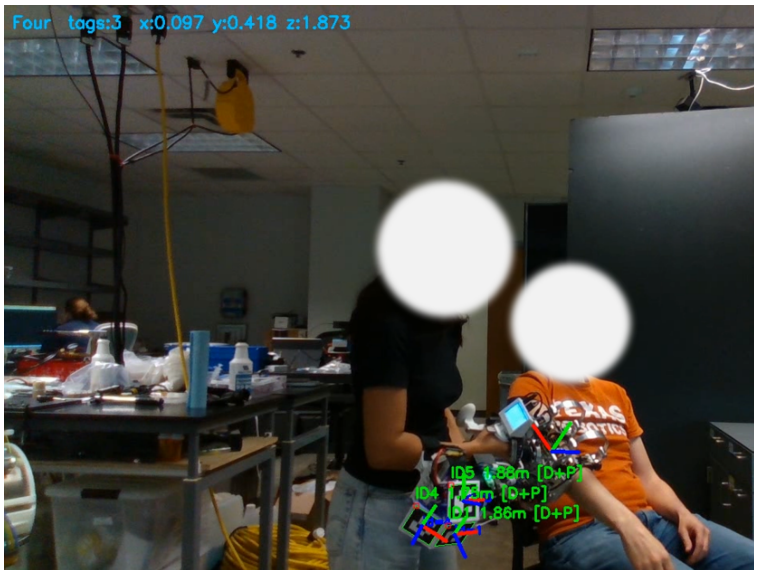


Figure 13: Illustration of the GUI implemented on the wearable data collection tool.

2. Run the *SMPL Human Detection Node* to generate the human pose rule. See Figure 14.

(a) Download the repo from GitHub: https://github.com/jidaxian010/humanpose.git

(b) Activate the conda environment:
```
conda env create -f environment.yaml
conda activate humanpose
```

(c) Select the same rosbag from the previous step and generate an MP4 preview:
```
python scripts/preview.py
```

(d) Manually crop out the approach and ending stage. Generate the human pose rulebook:
```
python scripts/generate_rule.py
```

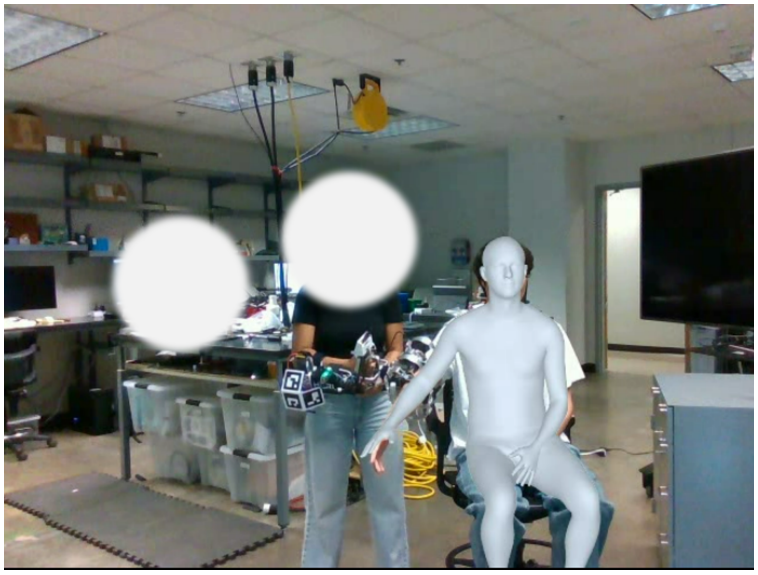


Figure 14: Illustration of the GUI implemented on the wearable data collection tool.

This development period for these algorithms lasted up to 4 weeks for all of the participants. Multiple Optimo control policies were generated in parallel so multiple participants could be scheduled within the same day. Once the policy was validated, participants were invited back for their optional second session.

3.5 PHASE 2 PROCEDURE

Upon the participant returning, the researcher reintroduced the robotic system and verified that the participant still met the inclusion criteria. Participants reviewed their signed consent forms, and after verbally confirming they wished to proceed, the session began.

The system operated under the constant supervision of the clinician and researcher and was immediately halted if an unexpected event occurred or if the participant indicated any discomfort. After all interactions were completed, the human subject data collection phase concluded. System data was successfully transferred to the encrypted UT Box, and local robotic memory was erased. The following procedure was followed to complete phase 2.

1. Position one camera to capture the participant motion.
    (a) The camera should directly face the human torso for human pose detection.
2. Attach the ANTERO Gripper to the Optimo arm with the two pins.
    (a) Plug in the power and communication connections and then power on the gripper
    (b) After the homing sequence, remove the safety stops (four of them).
3. Position the patient in the correct spot and set up the initial pose for the robot.

(a) Verify a test motion of the exercise, checking for human pose estimation consistency and accuracy.

(b) In `~/CODE/roboligent_sdk/optimo_controller/bin`, run `./quick_start` to bring up the Optimo robot from the GUI, enable the robot, and set the robot into free motion mode.

(c) Place the participant next to the Optimo robot

(d) handle/back-drive the robot so the gripper contacts the patient arm While the participant holds their arm in the proper starting orientation.

(e) Set this robot configuration as the initial pose and enable the Optimo brakes. Update:
`RobotPT/posEstimate/scripts/generate_gripper.py`
and regenerate the joint trajectory with the new initial pose.

(f) Stop the Optimo robot and kill the GUI.

4. Take off the wearable device and bring up the Optimo robot with *Optimo Bringup Node* (Terminal 1):
```
ros2 launch optimo_bringup optimo.launch.py
```

5. Run the external wrench baseline and record:

(a) Launch the *Safety Node* (Terminal 2):
```
ros2 launch optimo_safety_monitor safety_monitor.launch.py
```

(b) Start external wrench baseline recording (Terminal 3):
```
ros2 service call /optimo/safety_monitor/record_baseline std_srvs/srv/Trigger
```

(c) Play the trajectory (Terminal 4):
```
ros2 service call /optimo/optimo_effort_controller/play_traj_cb
optimo_msgs/srv/PlayTrajCb
"{goal: {filepath: 'RobotPT/posEstimate/data/p8-b2-g/p8-b2-g.trj',
param: {imp: {impedance: 7, max_force: 10, stop_sensitivity: 10},
recovery_speed: 2, accept_diff_start: true}}}"
```

(d) Save the external wrench baseline recording (Terminal 5) and move it to the desired data path:
```
ros2 service call /optimo/safety_monitor/save_baseline std_srvs/srv/Trigger
cp tmp/optimo_wrench_baseline.csv ~/CODE/RobotPT/posEstimate/data/p8-b2-g
```

(e) Kill all terminals.

6. Put the gripper back on the end effector on the Optimo robot.

7. Check data before run. Take p8-b2 (Participant 8, Exercise b2) for example. In `RobotPT/posEstimate/data/p8-b2-g`, we should have `force_rule.yaml`,

`optimo_wrench_baseline.csv`, and `p8-b2-g.trj`. In `HumanPose/data/p8-b2-h`, we should have `arm_vector_rules.yaml`.

8. If verified, start the following in sequence to launch the whole system.

   (a) *Optimo Bringup Node* (Terminal 1/A):
   `ros2 launch optimo_bringup optimo.launch.py`

   (b) Launch RealSense camera (Terminals B):
   `ros2 launch human_pose_vector pose_vector.launch.py`

   (c) Launch the *SMPL Human Pose Node* (Terminal C `bash ~/ros2_ws/run_pose_node.sh --ros-args -p rulebook:=HumanPose/data/p8-b2-h/arm_vector_rules.yaml -p alpha:=0.33 -p warning_alpha:=0.1`

   (d) Launch the *Safety Node* (Terminal D):
   `ros2 launch optimo_safety_monitor safety_monitor.launch.py baseline_file:=RobotPT/posEstimate/data/p8-b2-g/optimo_wrench_baseline.csv force_rules_file:=RobotPT/posEstimate/data/p8-b2-g/force_rule.yaml wrench_force_threshold:=20.0`

   (e) Run the textitNTERO Gripper Node (Terminal E):
   `cd ~/CODE/optimo_ws`
   `ros2 run antero_gripper antero_gripper_test_node --ros-args -p port:=/dev/ttyUSB0`

   (f) Start Recording (Terminal F):
   `ros2 bag record -a -o ~/Downloads/p8-b2-h`

9. (Optional) After the participant verbally acknowledges they are ready, we can do a mock run with gripper open to have the participant understand how the Optimo will move:

   (a) Play the trajectory:
   `ros2 service call /optimo/optimo_effort_controller/play_traj_cb optimo_msgs/srv/PlayTrajCb "{goal: {filepath: 'RobotPT/posEstimate/data/p8-b2-g/p8-b2-g.trj', param: {imp: {impedance: 7, max_force: 10, stop_sensitivity: 10}, recovery_speed: 2, accept_diff_start: true}}}"`
   (where data path and data filename may vary)

10. After the participant verbally acknowledges they are ready, replay the trajectory:

(a) In the ANTERO Gripper Node Terminal (Terminal E), press the space bar to close the gripper (must be in that terminal window).

(b) Play the trajectory (Terminal G):
`ros2 service call /optimo/optimo_effort_controller/play_traj_cb optimo_msgs/srv/PlayTrajCb "{goal: {filepath: 'RobotPT/posEstimate/data/p8-b2-g/p8-b2-g.trj', param: {imp: {impedance: 7, max_force: 10, stop_sensitivity: 10}, recovery_speed: 2, accept_diff_start: true}}}"`
(where data path and data filename may vary)

(c) Ready the press the Emergency Stop for the Optimo or gripper open key in case of emergency.

(d) Open the gripper when the trajectory is done

11. Kill all terminals.

12. Verify the integrity of the recorded bag files.

13. Repeat steps 6-12 until three trials have been completed.

14. Repeat steps 1-12 until both sets of exercises have been completed. The entire phase 2 process takes less than 40 minutes.

3.6 Post-Collection Procedure

Following data collection, both quantitative and qualitative data were analyzed. All electronic records were stored on secure, university-owned computers, while physical documents were maintained within the secure HCRL facility. De-identified data and corresponding results are summarized in the following sections.

We first performed an analysis at the individual participant level. For each participant, three experimental trials were collected for each motion condition (A and B). These trials were compared against the corresponding ground truth or command signals, including joint states, gripper states, interaction forces, and human pose estimates. For each modality, we computed the mean and standard deviation across the three trials to assess repeatability and consistency within each participant.

We then conducted a population-level analysis focusing on gripper pose (position and orientation: x, y, z, roll, pitch, yaw) and human pose (Cartesian positions). For each participant p and trial k, we first compute the trajectory-level error with respect to the ground truth:

$$E_{p,k} = \frac{1}{T} \sum_{t=1}^{T} \left\| x_p^{(k)}(t) - x_p^{\mathrm{GT}}(t) \right\|^2$$

where $x_p^{(k)}(t)$ denotes the measured signal at time t for trial k, and $x_p^{\mathrm{GT}}(t)$ denotes the corresponding ground truth.

We then compute the within-subject mean error:

$$\mu_p = \frac{1}{K} \sum_{k=1}^{K} E_{p,k}$$

where $K = 3$ is the number of trials per participant. Finally, the population-level statistics are obtained by averaging across all participants:

$$\mu_{\text{pop}} = \frac{1}{P} \sum_{p=1}^{P} \mu_p, \quad \sigma_{\text{pop}} = \sqrt{\frac{1}{P} \sum_{p=1}^{P} (\mu_p - \mu_{\text{pop}})^2}$$

where $P = 5$ is the total number of participants.

Finally, we evaluated safety-related performance metrics. Specifically, we computed the percentage of time during each trial in which the system entered a warning state, as determined by violations of either the force-based or human pose-based safety rules. In addition, we recorded the total number of trials that were terminated due to unsafe conditions triggered by these rules. These metrics were aggregated separately for motion conditions A and B to enable comparative analysis of safety performance.

4 RESULTS AND DISCUSSION

Ten participants were enrolled in this pilot study. All of the participants were between the ages of 23 and 45. Every participant was recruited from the Aerospace Engineering Building at the University of Texas at Austin. Every participant had at least one college-level degree in engineering. Participants 1-5 only participated in phase 1 and were used to refine the ARPTS before proceeding with the last 5 participants. Only data gathered with participants 6-10 are shown in the results of this report. Three clinicians were enrolled in the pilot study, two clinicians (C1 and C2) participated in phase 1 and two clinicians participated in phase 2 (C1 and C3). Clinicians C1 and C3 have DPTs and C2 is a physiotherapist.

Qualitative feedback on the three devices that make up the ARPTS were gathered from the participants and the clinicians using Likert-type scales. For example, the participants and the clinicians were asked how controllable the Optimo manipulator seemed, from 1/uncontrollable to 5/controllable. These questionnaires were administered promptly after finishing data collection for phase 2 of the pilot study. The responses from participants 6-10 and clinicians (C1, C2, and C3) are shown in Figure 15.

A notable limitation of this study is that the enrolled participants and clinicians may have introduced bias into the qualitative data, given their inherent familiarity with robotic devices stemming from their educational and professional backgrounds. Additionally, several functional issues concerning the ANTERO Gripper, the wearable device, and the Optimo robotic manipulator were identified during preliminary testing and the execution of the pilot study.

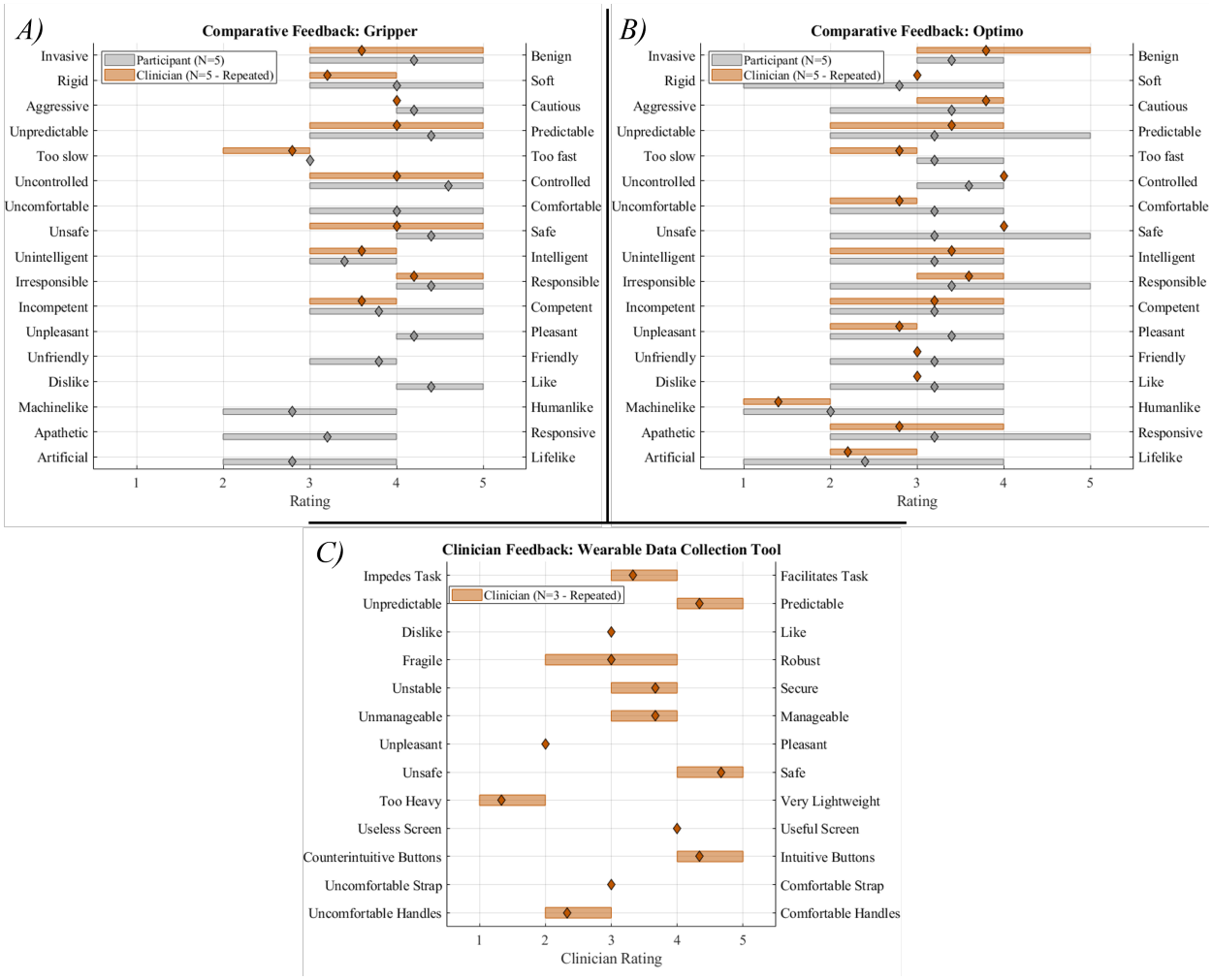


Figure 15: The results from the Likert-type scaled questionnaire given to the participants and the clinicians. The means (diamond markers) and ranges (boxes) of the responses to the Likert scale questionnaires. In general, the participants reported more positively than the clinicians (A and B). Since the participants in this pilot study did not directly interact with the WDCT, only the clinicians provided feedback on the wearable (C). In all of the Likert scales, a 5/5 is the best possible response.

Table 4: The means and standard deviations regarding the participants and clinicians' overall satisfaction with the ANTERO Gripper, Optimo Manipulator, and the WDCT.

	ANTERO Gripper	**Optimo Manipulator**	**Wearable Tool**
Participants	3.9 ± 0.8	3.1 ± 1.0	NA
Clinician	3.7 ± 0.7	3.1 ± 0.8	3.3 ± 1.1

4.1 THE ANTERO GRIPPER

Isolated testing conducted prior to the pilot study revealed several functional discrepancies in the ANTERO Gripper. Primarily, the measured grasp force produced by each finger differed significantly from the estimated grasp force, see Figure 16. The mean grasp force error was approximately -1.4 N; this negative value indicates that the force estimator reported higher forces than the finger actually exerted. The mean grasp force error also varies significantly with the grasp force magnitude, as can be seen in Figure 16C. Although this tendency provides a conservative safety margin for the ANTERO Gripper, the anticipated magnitude of the grasp force error was expected to be less than 0.5 N. This discrepancy is likely attributable to dimensional variations within the machined components, inaccuracies among the three encoders on each finger, or a combination of both. A summary of the median and the interquartile range (IQR) of the magnitude of the grasp force error from the five finger configurations tested are shown in Table 5.

The median and the interquartile range (IQR) of the magnitude of the grasp force error from the five finger configurations tested are shown in Table 5.

θ_1 (rad/deg)	θ_2 (rad/deg)	e_a (mm)	e_b (mm)	**Median Error** (N)	**Error IQR** (N)
1.05/60	0.79/45	68.4	76.4	-1.02	0.61
1.05/60	1.05/60	64.0	62.8	-2.04	1.54
1.05/60	1.31/75	60.5	52.8	-2.11	1.58
1.31/75	1.31/75	46.7	44.1	-0.96	1.40
1.57/90	1.57/90	34.6	30.5	-2.10	2.87

Table 5: Grasp force testing with five different finger positions. The grouped median error and error interquartile range are also shown in Figure 16D.

Because incremental encoders were utilized to measure the angular positions of the proximal and distal links of the ANTERO fingers, the system required homing each time the gripper was powered on. Multiple homing strategies were evaluated to ensure accuracy. These included a multi-variable regression applied as the fingers swept through their workspace, the use of a 3D-printed object paired with a force sensor to minimize the error between measured and estimated grasp forces, and the integration of 3D-printed inserts acting as hard stops for the compliant five-bar mechanism. Ultimately, utilizing hard stops coupled with manual tuning of the angle offsets—validated via regression and grasp force error minimization—proved to be the most accurate approach and was implemented. Furthermore, the incremental encoders featured a resolution of 7.5 arcminutes, which is relatively coarse compared to standard encoders used in robotic grippers. While these optical encoders are relatively inexpensive, lightweight, and impervious to electromagnetic interference, the low resolution likely exacerbated the grasp force error and reduced the overall grasp force resolution. For future iterations of the ANTERO Gripper, implementing higher-resolution absolute encoders on the proximal and distal links is highly recommended to mitigate force estimation errors.

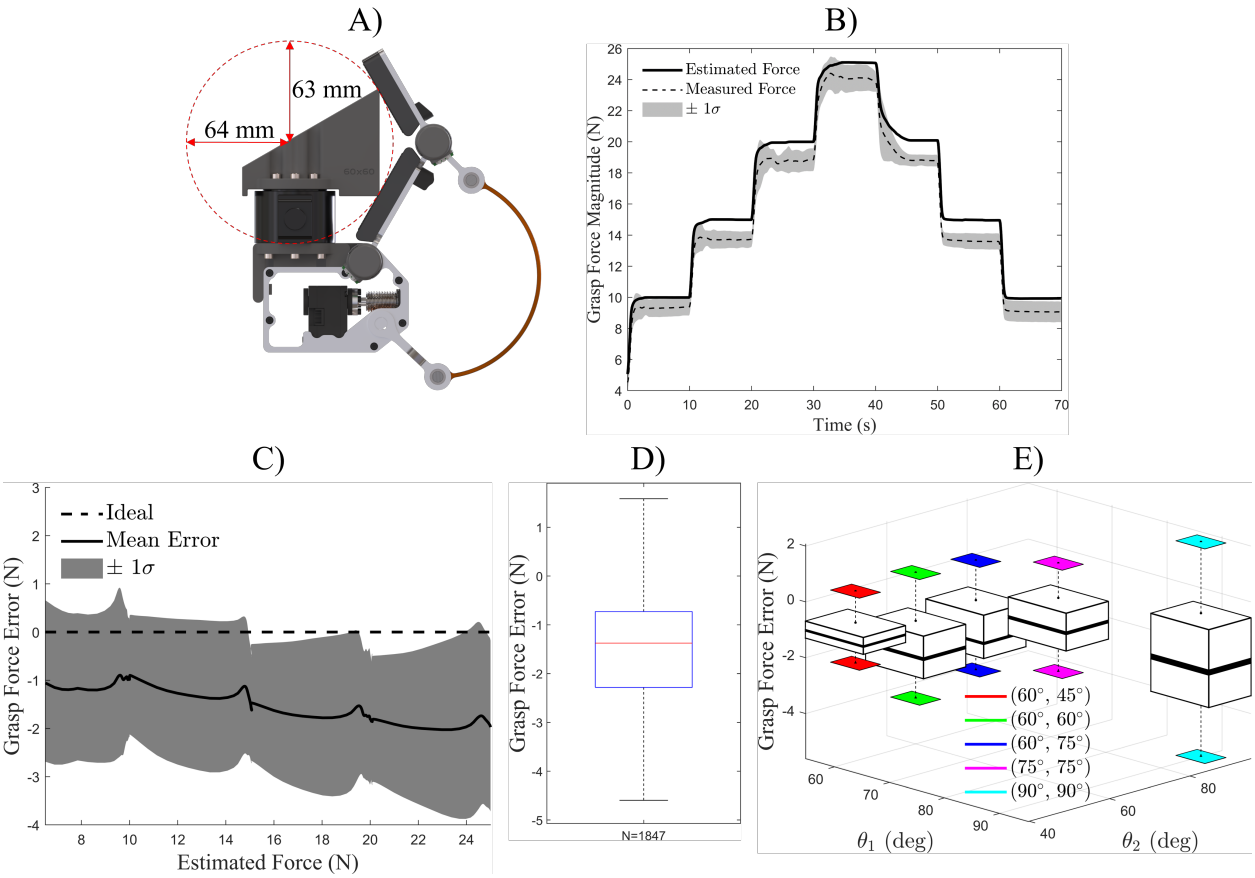


Figure 16: Results from testing the ANTERO Finger force estimator. A) Rendering of the test setup for the (60°, 60°) trials. B) Differences between the estimated and measured magnitude of the grasp forces. C) Estimation error vs the estimated force with a culmination of 25 trials tested at the five unique finger positions, θ_1 and θ_2 (in degrees). D) Grouped force estimation error over all the trials and force magnitudes. Generally, the median grasp force error is -1.37 Newtons and an interquartile range of 1.55 N. E) Differences between the estimated force error and the finger position. Here σ refers to the standard deviation of the data.

While the ANTERO Gripper benefits practically from relying solely on kinematic encoder data to estimate grasp force, substantial effort was required to tune and validate the estimator against actual measured forces. As an alternative, future designs could incorporate direct tactile sensing. The finger pads were intentionally designed with a thickness of 11 mm to accommodate various styles of tactile force sensors [42, 43]. Integrating these sensors would introduce sensory redundancy, thereby improving the overall safety and user comfort of the system.

Qualitative feedback yielded overall satisfaction scores of 3.9 ± 0.8 from participants and 3.7 ± 0.7 from clinicians. The lowest-rated aspects of the ANTERO Gripper (defined as

ratings *leq* 3) indicated that the device was slow, lacked responsiveness, and felt artificial or machine-like. In supplementary comments, Participant 9 noted that the gripper seemed to "impart more pressure than a human would." This observation directly correlates with a specific incident during this participant's trial, wherein an improper homing procedure for one of the fingers (operator error) caused the gripper to apply excess pressure to the participant's upper arm. The device was immediately disabled; although the participant experienced temporary discomfort, no injuries were sustained. Because this malfunction occurred during their final set of exercises, Participant 9's trial was terminated prematurely. To address the broader qualitative feedback and improve future clinical evaluations, subsequent designs should utilize a lower gear ratio to enhance both the operational speed and the frequency response of the fingers' motion and force output.

4.2 THE WEARABLE DATA COLLECTION TOOL

The wearable device proved to be the most contentiously debated subsystem within the ARPTS, even though the average clinician satisfaction score was 3.3 ± 1.1. The participating clinicians possess years of experience administering physical therapy bimanually. Because the wearable device constrained them to single-handed interactions with participants, the clinicians were forced to adapt their established therapeutic techniques. This constraint likely contributed to the relatively negative evaluations of the WDCT, with both clinicians reporting low scores for overall likability and comfort. Additionally, the clinicians indicated that the WDCT was excessively heavy, with a total mass of 3.9 kg (inclusive of the ANTERO Gripper and battery). Furthermore, both evaluating clinicians noted that "it was difficult to feel the force imparted on the participant," highlighting a critical lack of haptic feedback, something clinicians heavily rely on when treating patients.

Based on these findings, it is recommended that future clinical evaluations of the ARPTS reconsider the utilization of the WDCT. A viable alternative for data collection would involve the clinician wearing sensorized gloves to capture interaction force data, coupled with a motion capture system to track the kinematic positions of both the therapist and the patient. Utilizing this data, a motion re-targeting strategy could be implemented to generate the Optimo control policies. Finally, given that therapists typically rely on bimanual manipulation to adjust and support patient limbs, the integration of a dual-arm robotic system should also be considered for future iterations.

4.3 The Optimo Manipulator

The overall user satisfaction with the Optimo controller was rated as 3.1 ± 1.0 by participants and 3.1 ± 0.8 by clinicians, indicating moderate and consistent acceptance across both groups. Throughout all experimental trials, no unexpected events, safety violations, or hardware/software malfunctions were observed, demonstrating the reliability and robustness of the system.

As shown in Fig. 17, the commanded joint trajectories closely align with the actual exe-

cuted trajectories, indicating accurate low-level control of the manipulator. Task A exhibits better overall alignment compared to Task B. In particular, for Task A, noticeable bias is primarily observed in joints 2 and 4, whereas for Task B, deviations are more prominent in joints 1, 3, and 5. Despite these joint-specific discrepancies, the overall tracking performance remains stable across both tasks.

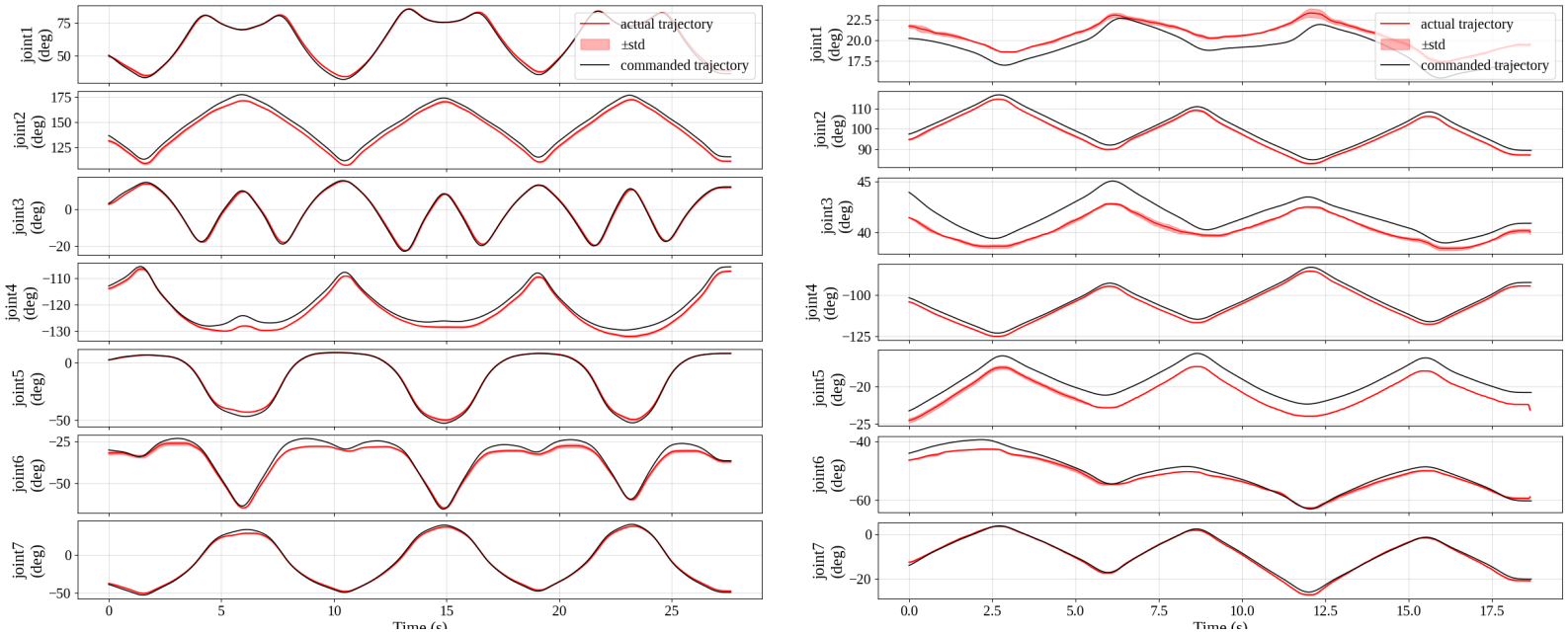


Figure 17: Joint trajectory comparison between Task A and Task B on one participant.

In Fig. 18, we compare the gripper pose trajectories obtained from the robot (actual) and motion capture system (ground truth), both expressed in the initial gripper frame. Compared to joint-space tracking, larger discrepancies are observed in task-space trajectories due to the accumulation of kinematic and sensing errors along the chain. Nevertheless, the overall trends remain consistent, indicating that the system maintains reasonable end-effector accuracy. Task A demonstrates better alignment with the ground truth, while Task B exhibits smaller variance and standard deviation, suggesting more consistent but slightly biased motion execution.

Fig. 19 illustrates the interaction forces measured at the end effector. The light gray region represents the warning zone, which triggers a visual warning, while the dark gray region indicates unsafe conditions that would result in an automatic stop command. Across both tasks, the measured forces remain largely within the predefined safe bounds. Occasional excursions into the warning zone are observed, but no measurements reach the unsafe region, confirming safe operation throughout all trials.

Finally, Fig. 20 presents the comparison between the observed human pose and the ground truth reference. Due to natural variability and unpredictability in human motion, the alignment is less precise compared to robot joint or gripper trajectories. However, the

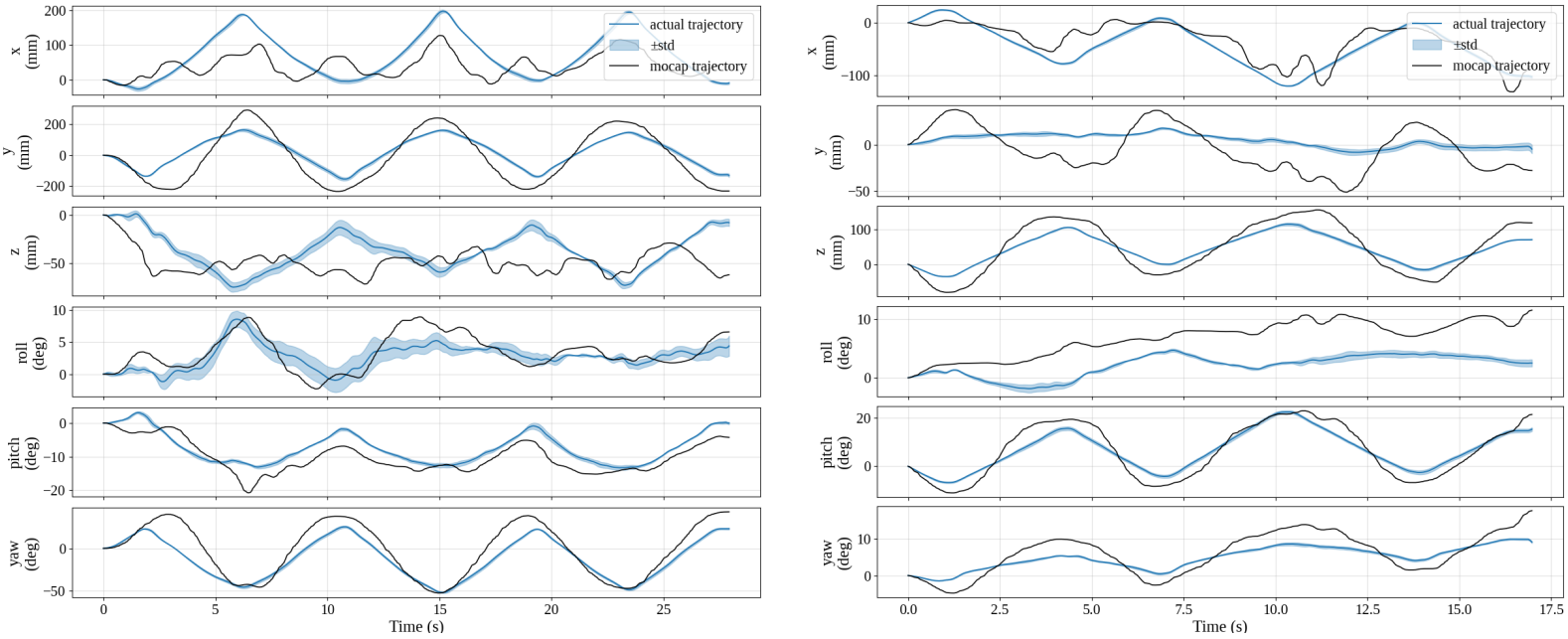


Figure 18: Gripper trajectory comparison between Task A and Task B on one participant.

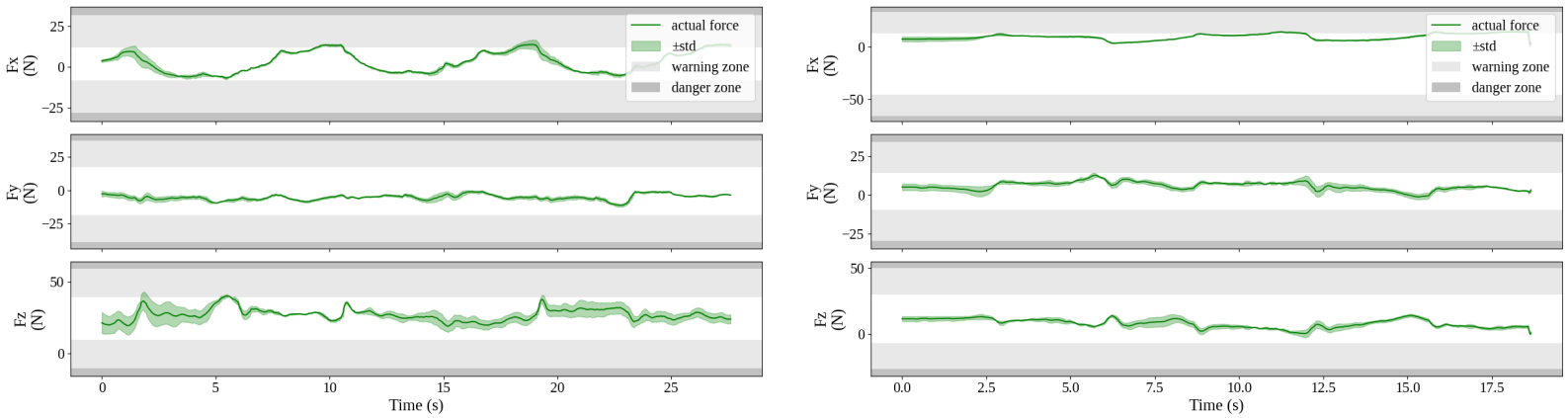


Figure 19: Force comparison between Task A and Task B on one participant.

overall motion patterns remain consistent with the reference, and all measured human poses stay within the predefined safe bounds, without entering even the warning zone. Additionally, Task B involves a smaller range of motion, which contributes to reduced variability in the observed human pose.

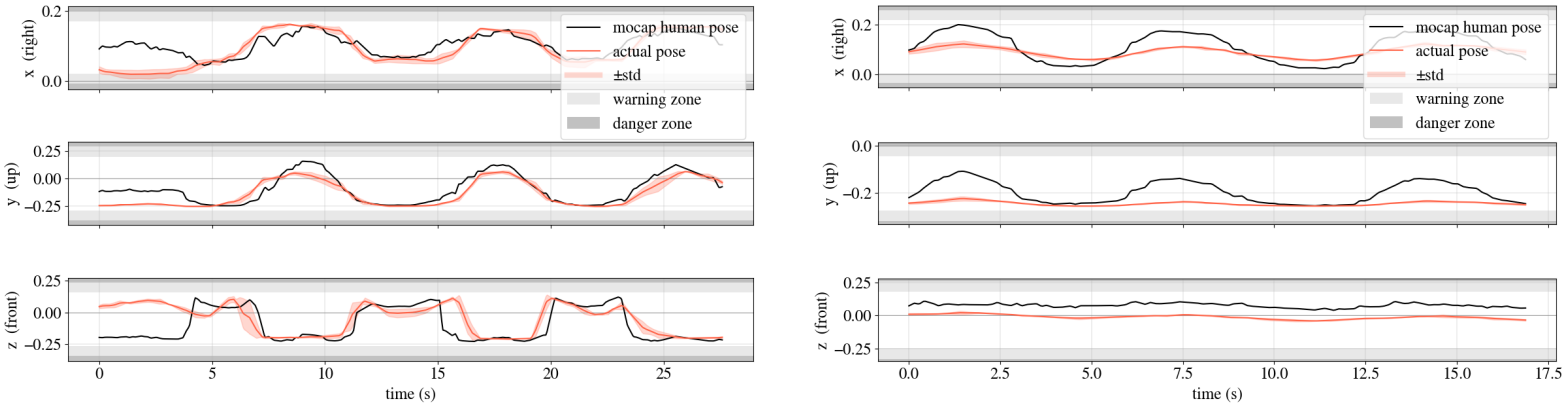


Figure 20: Human pose comparison between Task A and Task B on one participant.

Overall, these results demonstrate that the Optimo manipulator achieves accurate trajectory tracking, maintains safe interaction forces, and operates reliably in human-in-the-loop scenarios, while accommodating natural variability in human motion.

Table 6: Gripper pose analysis across all trials

	Mean RMSE		Std (Subj)		Within-Subj Std	
Task	A	B	A	B	A	B
x (mm)	41.95	42.55	22.35	28.50	4.031	11.19
y (mm)	68.74	48.99	31.79	47.64	7.187	11.49
z (mm)	35.33	52.06	6.915	26.59	4.313	13.96
roll (deg)	4.545	7.354	2.439	2.588	0.5742	2.315
pitch (deg)	4.604	8.960	1.182	4.597	0.3341	2.933
yaw (deg)	15.23	10.55	6.726	12.2	1.383	0.8751

For Table 6, Mean RMSE denotes the mean root-mean-square error across all participants and all trials, Std (Subj) denotes the standard deviation of that mean across participants and trials, and Within-Subj Std denotes the simple average of the per-subject standard deviation. Overall, the gripper pose errors are largest in the translational directions, especially along the y axis, where the Mean RMSE reaches 68.74 for Task A and 48.99 for Task B. In comparison,

the angular errors are smaller in magnitude, although yaw remains relatively large at 15.23 for Task A and 10.55 for Task B.

The relatively large Std (Subj) values, particularly for the translational components, indicate that gripper pose performance varied meaningfully across participants and trials. The Within-Subj Std values are smaller than the between-subject variability, suggesting that part of the spread comes from differences among participants rather than from inconsistency within a single participant alone. Taken together, these results suggest that the gripper pose estimates exhibit nontrivial error and appreciable variability, with translational accuracy being the dominant limitation.

Table 7: Human pose analysis across all trials

	Mean RMSE		Std (Subj)		Within-Subj Std	
Task	A	B	A	B	A	B
x	0.003335	0.04953	0.001421	0.008779	0.005976	0.005613
y	0.05539	0.06678	0.001542	0.008533	0.003864	0.005884
z	0.09882	0.1060	0.02263	0.01676	0.005621	0.005072

For Table 7, the same definitions apply: Mean RMSE is the average RMSE across all participants and all trials, Std (Subj) captures the variation of that mean across participants and trials, and Within-Subj Std is the simple average of each participant's standard deviation. The values in this table are small because they are computed from products of unit vectors derived from the human pose, so they reflect orientation relationships rather than translational position errors in physical units. Accordingly, the magnitudes should be interpreted as directional consistency metrics, not as distances. Refer to Equation 6 for human pose vector details.

The dispersion metrics are also comparatively small, which suggests that these orientation relationships are stable both across subjects and within individual subjects. Although the z direction shows the largest Mean RMSE among the three reported axes, the absolute magnitude remains low, indicating that the human pose estimation pipeline preserves these orientation relationships consistently across the tested conditions. Overall, the table suggests that the human pose representation is relatively robust and repeatable for both Task A and Task B.

For the safety summary, the warning duration ratio represents the proportion of task time during which the Safety Node issued a warning, while the stopped unsafe trial ratio represents the proportion of total trials that were classified as unsafe and actively halted. The force rule generated warnings for a substantially larger fraction of the task duration than the pose rule, with values of 30.7% and 32.8% for Tasks A and B, respectively, compared with 10.4% and 6.7% under the pose rule. This suggests that the force-based criterion was

Table 8: Safety rule summary across all trials

	Force Rule		Pose Rule	
Task	A	B	A	B
warning duration ratio (%)	30.7	32.8	10.4	6.7
stopped unsafe trial ratio (%)	0	11	0	0

triggered more frequently and behaved more conservatively during operation.

Despite the relatively frequent warnings, actual safety stops were rare. Only Task B under the force rule resulted in any stopped unsafe trials, at 11%, whereas all other task-rule combinations remained at 0%. This pattern suggests that many warnings served as precautionary alerts rather than escalating to full trial termination, and it further indicates that the force rule was the more restrictive safety mechanism in the tested scenarios.

References

[1] Veronika Pav, Xiaoning Yuan, Brad Isaacson, Courtney Colahan, and Ben Hando. Burden of musculoskeletal injuries in u.s. active duty service members: A 12-year study spanning fiscal years 2010-2021. *Military Medicine*, 189(Supplement_4):1-9, 2024.

[2] Brad M. Isaacson, Veronika Pav, Xiaoning Yuan, Ben Hando, Courtney Colahan, Phillip M. Newman, Amanda Edwards-Stewart, Tolulope Dada, Tanja Roy, Kathryn M. Taylor, and Keith G. Hauret. Lower extremity musculoskeletal injuries in united states active duty service members: Prevalence, incidence, health care utilization, and cost analysis spanning fiscal years 2016–2021. *Military Medicine*, 189(Supplement_4):22–32, 2024.

[3] Courtney Colahan, Xiaoning Yuan, Brad M. Isaacson, Veronika Pav, Ben Hando, Phillip M. Newman, Amanda Edwards-Stewart, Tolulope Dada, Tanja Roy, Kathryn M. Taylor, and Keith G. Hauret. Musculoskeletal head and neck injuries in united states active duty service members: Prevalence, incidence, health care utilization, and cost analysis spanning fiscal years 2016–2021. *Military Medicine*, 189(Supplement_4):56–65, 2024.

[4] Justin Dougherty, Sarah Walmsley, and Peter G. Osmotherly. Passive range of movement of the shoulder: A standardized method for measurement and assessment of intrarater reliability. *Journal of Manipulative and Physiological Therapeutics*, 38(3):218–224, 2015.

[5] Carlos Lavernia, Michele D'Apuzzo, Mark D. Rossi, and David Lee. Accuracy of knee range of motion assessment after total knee arthroplasty. *The Journal of Arthroplasty*, 23(6 Suppl):85–91, 2008.

[6] Jakob Kristensen and Andy Franklyn-Miller. Resistance training in musculoskeletal rehabilitation: a systematic review. *British Journal of Sports Medicine*, 46(10):719–726, 2012.

[7] Craig Liebenson and Curtis Thor Rigney. Manual resistance techniques in rehabilitation. In Sasha Chaitow and Sandy Fritz, editors, *Chaitow's Muscle Energy Techniques*, pages 273–290. Elsevier Health Sciences, 5 edition, 2023.

[8] Deirdre Cowper Ripley, Dean Reker, Jeffrey Hayes, Theresa Hurd, and Kathryn Hawley. Geographic access to vha rehabilitation for traumatic injured oef/oif veterans. VA Health Services Research & Development (HSR&D) Research Brief, 2009. Accessed: 2025-09-15.

[9] Michael Hearn, Christopher Pinto, and Jennifer L. Moss. Evaluating the connection between rural travel time and health: A cross-sectional analysis of older adults living in the northeast united states. *Journal of Primary Care & Community Health*, 15, 2024.

[10] Kalpana P. Padala, Kerrie B. Wilson, C. Heath Gauss, Jessica D. Stovall, and Prasad R. Padala. Va video connect for clinical care in older adults in a rural state during the covid-19 pandemic: Cross-sectional study. *Journal of Medical Internet Research*, 22(9):e21561, 2020.

[11] Steven Z. George, Rogelio A. Coronado, Julie M. Fritz, John D. Childs, Anthony Delitto, Charles W. MacDonald, and Jason M. Beneciuk. Group physical therapy for musculoskeletal conditions in military and civilian populations: A randomized clinical trial. *Journal of Orthopaedic & Sports Physical Therapy*, 54(1):1–34, 2024.

[12] Pablo Rodríguez Sánchez-Laulhé, Luis Gabriel Luque-Romero, Francisco José Barrero-García, Ángela Biscarri-Carbonero, Jesús Blanquero, Alejandro Suero-Pineda, and Alberto Marcos Heredia-Rizo. An exercise and educational and self-management program delivered with a smartphone app (carehand) in adults with rheumatoid arthritis of the hands: Randomized controlled trial. *JMIR mHealth and uHealth*, 10(4):e35462, 2022.

[13] Jan-Dierk Clausen, Niclas Nahen, Hauke Horstmann, Florian Lasch, Werner Krutsch, Christian Krettek, and Thomas Sanjay Weber-Spickschen. Improving maximal strength in the initial postoperative phase after anterior cruciate ligament reconstruction surgery: Randomized controlled trial of an app-based serious gaming approach. *JMIR Serious Games*, 8(1):e14282, 2020.

[14] Pablo Molina-Garcia, Marta Mora-Traverso, Rafael Prieto-Moreno, Andrea Díaz-Vásquez, Benny Antony, and Patrocinio Ariza-Vega. Effectiveness and cost-effectiveness

of telerehabilitation for musculoskeletal disorders: A systematic review and meta-analysis. *Annales de Physique et de Réadaptation Médicale*, 67(1):101791, 2024.

[15] Thomas Lens and Oskar von Stryk. Investigation of safety in human-robot-interaction for a series elastic, tendon-driven robot arm. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 4309–4314, 2012.

[16] G. Salvietti, Z. Iqbal, I. Hussain, D. Prattichizzo, and M. Malvezzi. The co-gripper: A wireless cooperative gripper for safe human robot interaction. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4576–4581, 2018.

[17] Iñaki Maurtua, Aitor Ibarguren, Johan Kildal, Loreto Susperregi, and Begoña Sierra. Human–robot collaboration in industrial applications: Safety, interaction and trust. *International Journal of Advanced Robotic Systems*, 14(4), 2017.

[18] Stephen N. Housley, Kathleen Fitzgerald, and Andrew J. Butler. Telerehabilitation robotics: Overview of approaches and clinical outcomes. In Roberto Colombo and Vittorio Sanguineti, editors, *Rehabilitation Robotics*, pages 333–346. Academic Press, 2018.

[19] Juan C. Moreno, Joana Figueiredo, and José L. Pons. Exoskeletons for lower-limb rehabilitation. In Roberto Colombo and Vittorio Sanguineti, editors, *Rehabilitation Robotics*, pages 89–99. Academic Press, 2018.

[20] Tomonori Yokoo, Mitsuyuki Yamada, Sho Sakaino, Shigeru Abe, and Toshiaki Tsuji. Development of a physical therapy robot for rehabilitation databases. In *2012 12th IEEE International Workshop on Advanced Motion Control (AMC)*, pages 1–6, 2012.

[21] J. Micah Prendergast, Stephan Balvert, Tom Driessen, Ajay Seth, and Luka Peternel. Biomechanics aware collaborative robot system for delivery of safe physical therapy in shoulder rehabilitation. *IEEE Robotics and Automation Letters*, 6(4):7177–7184, 2021.

[22] James Patton, David A. Brown, Michael Peshkin, Juan J. Santos-Munné, Alex Makhlin, Elizabeth Lewis, J. Edward Colgate, and David Schwandt. Kineassist: Design and development of a robotic overground gait and balance therapy device. *Topics in Stroke Rehabilitation*, 15(2):131–139, 2008.

[23] Francisco J. Ruiz-Ruiz, Jorge Ventura Gil, Cristina Urdiales, and Jesús M. Gómez-de Gabriel. Compliant gripper with force estimation for physical human–robot interaction. *Mechanism and Machine Theory*, 178:105062, 2022.

[24] F. Lara, J. Castellanos-Ramos, A. Trujillo-León, C. Urdiales, J. L. Benavides, and J. M. Gómez-de Gabriel. phri gripper with pressure sensing. In L. Marques, C. Santos, J. L. Lima, D. Tardioli, and M. Ferre, editors, *Robot 2023: Sixth Iberian Robotics Conference*,

volume 978 of *Lecture Notes in Networks and Systems*, pages 388–399, Cham, 2024. Springer.

[25] Joaquín Ballesteros, Francisco Pastor, Jesús M. Gómez-de Gabriel, Juan M. Gandarias, Alfonso J. García-Cerezo, and Cristina Urdiales. Proprioceptive estimation of forces using underactuated fingers for robot-initiated phri. *Sensors*, 20(10):2863, 2020.

[26] Zhiqiang Zhao, Xiaodong Li, Cheng Lu, Xin Liu, and Aiguo Song. Compliant manipulation method for a nursing robot based on physical structure of human limb. *Journal of Intelligent & Robotic Systems*, 100:973–986, 2020.

[27] S. L. Müller, S. Stiehm, S. Jeschke, and A. Richert. Subjective stress in hybrid collaboration. In A. Kheddar et al., editors, *Social Robotics. ICSR 2017*, volume 10652 of *Lecture Notes in Computer Science*, pages 599–608, Cham, 2017. Springer.

[28] Peter Chemweno, Liliane Pintelon, and Wilm Decre. Orienting safety assurance with outcomes of hazard analysis and risk assessment: A review of the iso 15066 standard for collaborative robot systems. *Safety Science*, 129:104832, 2020.

[29] Jule Bessler, Gerdienke B. Prange-Lasonder, Leendert Schaake, José F. Saenz, Catherine Bidard, Irene Fassi, Marcello Valori, Aske Bach Lassen, and Jaap H. Buurke. Safety assessment of rehabilitation robots: A review identifying safety skills and current knowledge gaps. *Frontiers in Robotics and AI*, 8:602878, 2021.

[30] Hongliang Hua, Zhenqiang Liao, and Jingbo Zhao. Design, analysis, and experiment of an underactuated robotic gripper actuated by linear series elastic actuator. *Journal of Mechanisms and Robotics*, 15(2):021002, 06 2022.

[31] Paolo de Leva. Adjustments to zatsiorsky-seluyanov's segment inertia parameters. *Journal of Biomechanics*, 29(9):1223–1230, 1996.

[32] Zachary Merrill, Subashan Perera, April Chambers, and Rakié Cham. Age and body mass index associations with body segment parameters. *Journal of Biomechanics*, 88:38–47, 2019.

[33] R. E. Anakwe, J. S. Huntley, and J. E. McEachan. Grip strength and forearm circumference in a healthy population. *Journal of Hand Surgery (European Volume)*, 32(2):203–209, 2007.

[34] L. M. C. Pereira, M. F. C. de Souza, F. J. Aidar, M. Getirana-Mota, A. M. de Santos-Junior, M. F. D. da S. Filho, M. A. Almeida-Santos, R. M. S. Rocha, R. R. de Almeida, L. Baumworcel, et al. Wrist circumference cutoff points for determining excess weight levels and predicting cardiometabolic risk in adults. *International Journal of Environmental Research and Public Health*, 21:549, 2024.

[35] M. C. Gonzalez, A. Mehrnezhad, N. Razaviarab, T. G. Barbosa-Silva, and S. B. Heymsfield. Calf circumference: Cutoff values from the nhanes 1999–2006. *The American Journal of Clinical Nutrition*, 113(6):1679–1687, 2021.

[36] Margaret A. McDowell, Cheryl D. Fryar, Cynthia L. Ogden, and Katherine M. Flegal. Anthropometric reference data for children and adults: United states, 2003–2006. Technical Report 10, National Center for Health Statistics (NCHS), Centers for Disease Control and Prevention, 2008.

[37] Chao-lei Chen, Lin Liu, Jia-yi Huang, Yu-ling Yu, Geng Shen, Kenneth Lo, Yu-qing Huang, and Ying-qing Feng. Thigh circumference and risk of all-cause, cardiovascular and cerebrovascular mortality: A cohort study. *Risk Management and Healthcare Policy*, 13:1977–1987, 2020.

[38] Erik Oberg, Franklin D. Jones, Holbrook L. Horton, and Henry H. Ryffel. *Machinery's Handbook*. Industrial Press, South Norwalk, CT, 30th edition, 2016.

[39] Andrea Calanca, Riccardo Muradore, and Paolo Fiorini. Impedance control of series elastic actuators: Passivity and acceleration-based control. *Mechatronics*, 47:37–48, 2017.

[40] Nicholas Paine, Sehoon Oh, and Luis Sentis. Design and control considerations for high-performance series elastic actuators. *IEEE/ASME Transactions on Mechatronics*, 19(3):1080–1091, 2014.

[41] Ye Zhao, Nicholas Paine, Steven Jens Jorgensen, and Luis Sentis. Impedance control and performance measure of series elastic actuators. *IEEE Transactions on Industrial Electronics*, 65(3):2817–2827, 2018.

[42] Tetsuya Narita, Satoko Nagakari, William Conus, Toshimitsu Tsuboi, and Kenichiro Nagasaka. Theoretical derivation and realization of adaptive grasping based on rotational incipient slip detection. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pages 531–537, 2020.

[43] Jiandong Xu, Jiong Pan, Tianrui Cui, Sheng Zhang, Yi Yang, and Tian-Ling Ren. Recent progress of tactile and force sensors for human–machine interaction. *Sensors*, 23(4), 2023.